

Phase 3

Spatiotemporal Functional Genomics

Phase 3A - Abstract and Methods Summary

Abstract-Ready Phase 3 Summary

Phase 3 was organized into two analytical parts to determine whether the molecular

mechanisms identified in Phase 2 have the spatial opportunity to operate within the human visual system. Part I — retinal and ocular localization combined adult whole-tissue screening with adult human retinal cell-type analysis. Bulk retinal data showed the strongest expression among affected protein-coding loci for PRPF8, SERPINF1, PITPNA, RPA1, INPP5K, and TLCD2. Cell-type resolution then identified the clearest spatial convergence in retinal pigment epithelium, where INPP5K, PITPNA, and SERPINF1 converge on phosphoinositide regulation, phospholipid handling, and PEDF-associated trophic/lipid signaling. PRPF8 was broadly represented across neural retina, including photoreceptors, while TLCD2 showed a preliminary cone-photoreceptor signal. In contrast, the predicted WDR81-RILP trafficking pair was not demonstrated as a shared adult-retinal neuronal module. Part II — central visual-relay

specificity tested whether the affected interval preferentially mapped onto magnocellular,

parvocellular, or koniocellular transcriptional programs in the human dorsal lateral geniculate nucleus. None of the 17 affected loci appeared among the reproducible human M-versus-P marker genes or principal K-cell markers reported in the reference dataset. Phase 3 therefore provides evidence about where affected loci are expressed, which visual-system cell

populations can plausibly experience their molecular effects, and whether the deletion preferentially maps onto a specific visual relay stream. The evidence to date supports a

distributed retinal and neural functional-reserve model, with the strongest cell-specific signal centered on RPE membrane/lipid biology, but does not support a selectively magnocellular lesion or establish a definitive visual-neurological phenotype.

What Phase 3 Provides

Phase 2 established mechanistic plausibility by determining what the affected loci

normally do. Phase 3 adds spatial opportunity: it tests whether those loci are normally active in the tissues and cell populations required for the proposed mechanisms to matter in the human visual system. Through the work completed to date, Phase 3 distinguishes broad retinal expression from cell-specific localization, tests whether molecularly related deleted genes are actually co-located in the same retinal compartment, and directly evaluates whether the deletion preferentially corresponds to one classical thalamic visual pathway. It therefore narrows the hypothesis without treating expression as proof of dysfunction.

Step-by-Step Method Used Through Phase 3D

Step 1 — Freeze the candidate set and hypotheses before examining expression results

All 17 structurally affected loci identified in Phase 1 were retained. No gene was removed because Phase 2 had assigned it weak visual or neurological relevance. The primary

Phase 3 hypotheses were frozen in advance, including membrane/endosomal co-localization,

direct retinal localization, WDR81-RILP overlap, developmental enrichment, magnocellular specificity, and regulatory overlap. This was done to reduce phenotype-driven selection and confirmation bias.

Step 2 — Separate loci according to assay compatibility

Protein-coding genes were treated as generally measurable by conventional bulk and single-cell RNA sequencing. PITPNA-AS1 and MIR22HG were retained with the recognition that lncRNAs can be poorly captured in some single-cell platforms. Mature MIR22 was not interpreted from ordinary poly(A)-selected RNA-seq because mature microRNAs require small-RNA or miRNA-specific measurement. RN7SL105P was treated cautiously because pseudogene homology can cause ambiguous short-read mapping.

Step 3 — Screen adult whole-body tissue expression

Adult expression resources including GTEx and NCBI Gene summaries were used to determine whether each affected protein-coding locus was broadly expressed, brain biased, peripheral-tissue enriched, or strongly tissue specific. This established general biological context before any retina-specific ranking was performed. Cross-platform expression magnitudes were not treated as directly interchangeable.

Step 4 — Perform the initial adult human retinal screen

Human ocular resources, especially eyeIntegratation and the Human Protein Atlas retinal series, were used to determine which affected genes were detectably expressed in adult retina. The strongest bulk retinal signals among the affected protein-coding genes were PRPF8, SERPINF1, PITPNA, RPA1, INPP5K, and TLCD2. Genes showing weak whole-retina abundance were not discarded because bulk measurements can dilute expression from rare cell populations.

Step 5 — Resolve the retinal signal to individual cell classes

Large human retinal single-cell atlases and independent adult human retinal datasets were used to test localization among photoreceptors, bipolar cells, amacrine cells, horizontal cells,

retinal ganglion cells, Müller glia, astrocytes, microglia, vascular populations, and RPE.

Published marker frameworks were used as quality-control references for identifying major retinal populations.

Step 6 — Test whether the Phase 2 molecular convergence survives cell-type resolution

The pre-specified membrane/endosomal candidate set was examined for shared localization. The strongest supported adult-retinal convergence became an RPE-centered **INPP5K-PITPNA-SERPINF1 module. PRPF8 was broadly present across neural retina, and**

TLCD2 showed a preliminary cone-associated signal. Conversely, WDR81 and RILP were not demonstrated as a shared mature retinal neuronal module. Negative results were retained rather than redefining the candidate set after inspection.

Step 7 — Move from retina to the central visual relay

Human dorsal lateral geniculate nucleus single-nucleus RNA-seq data from the Allen/Bakken comparative LGN resource were used to test the pre-specified magnocellular hypothesis. The analysis treated magnocellular, parvocellular, and koniocellular identities symmetrically rather than defining magnocellular enrichment as the desired result.

Step 8 — Compare affected loci with published human M/P/K transcriptional markers

The human dLGN study showed that M and P projection neurons largely form one transcriptomic MP population with a graded M-to-P continuum. The published reproducible M/P marker set and principal K-cell markers were compared against all 17 affected loci. None of the affected loci was a defining M, P, or K marker. The result was therefore recorded as a meaningful negative finding: current adult human dLGN transcriptomics does not support preferential magnocellular targeting by the deletion.

Step 9 — Update the model without converting expression into causality

The evidence from the retinal and dLGN analyses was integrated with the Phase 2 mechanisms. Expression was interpreted only as opportunity for a mechanism, not proof that the deletion produces dysfunction. The working visual-system model therefore shifted away from a selectively magnocellular lesion toward a distributed retinal and neural functional-

reserve model, with the strongest present spatial convergence in RPE. No definitive phenotype

was assigned.

Current Phase 3 Interpretation

Through Phase 3D, the study has established that several molecularly important deleted genes are normally positioned within biologically relevant human visual-system compartments. The strongest cell-type-specific convergence is INPP5K + PITPNA + SERPINF1 in RPE, while PRPF8 has broad neural-retinal availability and TLCD2 may contribute a cone-associated lipid component. At the same time, the data weaken two earlier possibilities: WDR81 and RILP have not been demonstrated as a shared adult-retinal trafficking module, and none of the affected loci demonstrates published magnocellular-specific transcriptional enrichment in adult human dLGN. The Phase 3 evidence therefore supports continued investigation of a distributed retinal/neural mechanism while rejecting, at the present level of evidence, a specifically magnocellular transcriptomic explanation.

Phase 3 remains in progress. Human V1 and developmental brain/retinal analyses have

not yet been completed and must not be represented as results.

Phase 3B - Research Through Phase 3D

Document scope. This document reproduces the Phase 3 research conducted through Phase 3D. The wording of the research sections is preserved; prior hyperlink-style source notes

have been converted to MLA-style parenthetical citations. Phase 3B was defined in the frozen protocol but was not executed as a separate results section before Phase 3C, so its protocol language is included with an explicit scope note rather than with invented results.

Phase 3A — Adult Whole-Tissue and Initial Ocular Expression Screen

1. Objective

Phase 3A asks whether the 17 loci affected by the 17p13.3 microdeletion are

transcriptionally active in adult human tissues relevant to the project's visual-neurological hypothesis.

The purpose is not to infer disease from expression.

Instead, Phase 3A establishes the first requirement for biological plausibility:

A deleted locus must normally be expressed in the relevant tissue before reduced dosage can plausibly perturb that tissue through its canonical
molecular function.

All 17 loci remain under consideration.

2. Data Sources

The current GTEx adult bulk-expression release is GTEx Analysis V11. The portal provides gene-level TPM and median tissue-expression matrices, including the current V11 median gene-level TPM file. ("GTEx Portal")

For ocular-specific analysis, eyeIntegration v2.12 integrates publicly available healthy human ocular RNA-seq datasets with a substantial subset of GTEx samples processed through a common pipeline. It provides downloadable gene- and transcript-level raw-count matrices and includes multiple ocular tissue categories. ("eyeIntegration")

The initial retinal screen below additionally uses current Human Protein Atlas retinal RNA-expression pages. These provide a 105-sample adult human retinal series and consensus normalized transcript measurements. These values are useful for screening but will not substitute for the independent eyeIntegration and single-cell replication planned later in Phase 3.

Current NCBI Gene expression summaries were used as an independent whole-body localization check.

3. Whole-Body Expression Pattern

The first result is that the deleted interval does not consist predominantly of retinaspecific or brain-specific genes.

Most affected protein-coding genes are broadly or ubiquitously expressed.

This is methodologically important.

A visual-neurological mechanism, if present, will therefore probably depend on:

cell type + developmental context + dosage sensitivity

rather than simply: brain-specific gene expression.

4. Whole-Tissue Classification

Gene

Current broad tissue pattern

Phase 3A interpretation

Ubiquitous; NCBI highlights adipose

Broadly available molecular

INPP5K

and lung

mechanism

Ubiquitous; brain is among highest-

PITPNA

Strong initial neural localization signal

expression tissues

Broad; strongest in placenta and

Primarily systemic/epithelial bulk

SLC43A2

kidney

pattern

Broad; enriched toward

Supports immune/endothelial rather

SCARF1

spleen/lung/lymphoid-endothelial

than primary neuronal axis

tissues

Bulk tissue does not discriminate

RILP

Ubiquitous

neural function

Expected for core spliceosomal

PRPF8

Ubiquitous

protein

TLCD2

Broad; adipose and colon prominent

No bulk brain enrichment

Broad in current expression summary,

Requires cell-type and regional

WDR81

although RefSeq specifically describes

resolution

predominant brain expression

Strongly biased toward liver and

Supports systemic fibrinolytic rather

SERPINF2

kidney

than neural role

Systemic expression does not preclude

SERPINF1

Broad, very high in liver/gallbladder

strong retinal expression

Ubiquitous, generally modest

SMYD4

No major adult tissue specificity

expression

Consistent with fundamental genome-

RPA1

Ubiquitous

maintenance role

Broad but brain is the highest reported

RTN4RL1

Strong adult neural localization signal

human tissue

Representative current NCBI values include PITPNA at approximately 31.8 RPKM in

brain, RTN4RL1 at approximately 5.0 RPKM in brain, SLC43A2 at approximately 24.5

RPKM in placenta and 14.0 in kidney, and SERPINF2 at approximately 275.1 RPKM in

liver

and 42.8 in kidney. ("PITPNA Gene"; "RTN4RL1 Gene"; "SLC43A2 Gene"; "SERPINF2 Gene")

INPP5K, RILP, PRPF8, TLCD2, SMYD4, and RPA1 are currently described as ubiquitous or broadly expressed. ("INPP5K Gene"; "RILP Gene"; "PRPF8 Gene"; "TLCD2 Gene"; "SMYD4 Gene"; "RPA1 Gene")

SCARF1 shows a lymphoid/endothelial-biased pattern, while SERPINF1 remains broadly expressed despite very strong expression in several peripheral tissues. ("SCARF1 Gene"; "SERPINF1 Gene")

WDR81 is an interesting discrepancy: the current RefSeq functional summary calls it **predominantly expressed in brain, whereas NCBI's aggregated expression line labels it**

ubiquitous. This should not be forcibly reconciled; instead it becomes a reason to inspect regional and cell-level brain datasets directly. ("WDR81 Gene")

5. Adult Human Retina Screen

The current Human Protein Atlas retinal series provides an initial quantitative screen for the 13 protein-coding genes.

Table 3A.1 — Adult Human Retina Expression

Gene

Retina RNA expression

Initial classification

PRPF8

76.4 nTPM

High

SERPINF1

56.9 nTPM

High

PITPNA

43.9 nTPM

High

RPA1

20.8 nTPM

Moderate-high

INPP5K

14.7 nTPM

Moderate

TLCD2

10.1 nTPM

Moderate

RTN4RL1

7.9 nTPM

Moderate-low

SMYD4

7.2 nTPM

Moderate-low

SLC43A2

1.3 nTPM

Low

WDR81

0.7 nTPM

Low in whole-retina bulk

SERPINF2

0.5 nTPM

Very low

Gene**Retina RNA expression****Initial classification****RILP****0.4 nTPM**

Very low in whole-retina bulk

SCARF1**0.2 nTPM**

Very low

Current source values are directly reported by the Human Protein Atlas retinal pages.

("PRPF8 Retina"; "SERPINF1 Retina"; "PITPNA Retina"; "RPA1 Retina"; "INPP5K Retina"; "TLCD2 Retina"; "RTN4RL1 Retina"; "SMYD4 Retina"; "SLC43A2 Retina"; "WDR81 Retina"; "SERPINF2 Retina"; "RILP Retina"; "SCARF1 Retina")

These values should be interpreted as screening measurements, not as direct measures of functional importance.

6. First Major Phase 3 Result: PRPF8 Is Strongly Represented in Adult Retina

PRPF8 is the highest-expressed affected protein-coding gene in the initial retinal screen at approximately:

76.4 nTPM.

("PRPF8 Retina")

This substantially strengthens the Phase 2 observation that PRPF8 is biologically relevant to retinal tissue.

However, the significance is specific:

PRPF8 has both strong retinal expression and independently established

retinal disease biology.

It still does not establish that whole-gene heterozygous deletion produces the same mechanism as known pathogenic PRPF8 variants.

7. SERPINF1 Is Also Strongly Expressed in Retina

SERPINF1 is expressed at approximately:

56.9 nTPM

in the adult retinal series. ("SERPINF1 Retina")

This is particularly meaningful because Phase 2 independently established PEDF as a:

- retinal neurotrophic factor;
- antiangiogenic factor;
- regulator of retinal survival;
- participant in PEDF-R-linked phospholipid signaling.

The Phase 3A result therefore provides direct spatial opportunity for the Phase 2 mechanism.

SERPINF1 is not merely a systemic secreted gene with incidental retinal literature.

It is substantially expressed within adult retinal tissue.

8. PITPNA Provides a Strong Link Between the Phase 2 Lipid Cluster and Retina

PITPNA is expressed at approximately:

43.9 nTPM

in adult retina. ("PITPNA Retina")

NCBI also identifies brain as one of PITPNA's highest-expression tissues. ("PITPNA Gene")

This makes PITPNA one of the first genes for which Phase 3 supports both:

adult neural expression

and: substantial retinal expression.

Because PITPNA was already one of the strongest Phase 2 phosphoinositide-homeostasis candidates, this finding materially strengthens the membrane-lipid hypothesis.

9. INPP5K Is Clearly Present in Adult Retina

INPP5K averages approximately:

14.7 nTPM

in the retinal series. ("INPP5K Retina")

More importantly, current Human Protein Atlas single-cell annotations classify INPP5K as group enriched in retinal pigment epithelial cells and late spermatids, providing an early indication that its retinal bulk expression may contain a strong RPE component. ("INPP5K Single Cell")

This observation is significant because it begins to localize one of the central phosphoinositide enzymes from Phase 2 to a specific ocular cell population.

It will require confirmation in the dedicated retinal single-cell phase.

10. RPA1 Is Substantially Expressed in Adult Retina

RPA1 shows approximately:

20.8 nTPM

with relatively limited variation across the 105 retinal samples. ("RPA1 Retina")

This fits its ubiquitous genome-maintenance role.

The result establishes that the tissue contains substantial RPA1 transcript and therefore that the Phase 2 haploinsufficiency mechanism is molecularly available in retinal cells.

It does not establish that retina is preferentially vulnerable to RPA1 loss.

11. RTN4RL1 Is Detectable in Retina and Brain

RTN4RL1 shows approximately:

7.9 nTPM

in adult retina. ("RTN4RL1 Retina")

Its whole-body pattern is additionally brain biased, with NCBI listing brain as its highest reported human tissue. ("RTN4RL1 Gene")

This supports the Phase 2 interpretation of RTN4RL1 as a legitimate neural/visualpathway candidate rather than a receptor whose optic-nerve biology is disconnected from human tissue expression.

The next decisive question is:

Which retinal population expresses RTN4RL1?

If it localizes primarily to retinal ganglion cells, that would align closely with the opticnerve regeneration mechanism described in Phase 2.

If it instead localizes to another population, the working model must be revised accordingly.

12. TLCD2 Is Detectable at Moderate Levels

TLCD2 shows approximately:

10.1 nTPM

in adult retina. ("TLCD2 Retina")

This is an important result because TLCD2 had weak direct retinal evidence in Phase 2.

Phase 3A therefore upgrades one specific proposition:

The TLCD2 phospholipid-composition mechanism has an opportunity to

operate in adult retinal tissue.

It does not yet establish which retinal cells express TLCD2 or whether one-copy reduction meaningfully alters retinal membrane composition.

13. WDR81 and RILP Produce an Important Bulk-Tissue Warning

WDR81 and RILP were among the strongest neural trafficking candidates in Phase 2.

Yet whole-retina expression is low:

WDR81 $\approx$ 0.7 nTPM

RILP $\approx$ 0.4 nTPM.

("WDR81 Retina"; "RILP Retina")

This does not currently falsify the WDR81–RILP hypothesis.

Instead, there are at least three possibilities:

1. They are genuinely weakly expressed in adult retina.

That would weaken a retinal mechanism.

2. They are expressed in a small retinal population.

Bulk retinal RNA would dilute such a signal.

3. Their principal relevance lies outside retina, for example in dLGN, cortex, developmental

progenitors, or other neural tissues.

Phase 3C–G will discriminate among these possibilities.

This is exactly why bulk tissue analysis was performed before cell-type interpretation.

14. SCARF1 and SERPINF2 Behave as Predicted from Phase 2

SCARF1 and SERPINF2 show only approximately:

0.2 nTPM

and:

0.5 nTPM

respectively in adult retina. ("SCARF1 Retina"; "SERPINF2 Retina")

Their whole-body patterns are much more compatible with:

SCARF1 → immune/endothelial biology

and: SERPINF2 → hepatic/systemic fibrinolysis.

This is a useful negative result.

Phase 3A currently provides no reason to elevate either gene as a direct retinal candidate.

They remain in subsequent unbiased analyses but should not receive retinal priority merely because they fall inside the deletion.

15. SLC43A2 Remains Weak as a Direct Retinal Candidate

SLC43A2 shows only approximately:

1.3 nTPM

in adult retina. ("SLC43A2 Retina")

Combined with its placenta/kidney-biased whole-body pattern, this further supports its

Phase 2 classification as primarily an epithelial nutrient-transport gene rather than a leading

direct visual-system candidate. ("SLC43A2 Gene")

A specialized retinal cell population could still express it, so it remains in the single-cell analysis.

16. SMYD4 Is Detectable but Not Retinally Exceptional

SMYD4 averages approximately:

7.2 nTPM

in adult retina. ("SMYD4 Retina")

Its whole-body expression is otherwise ubiquitous and relatively modest. ("SMYD4 Gene")

Thus Phase 3A establishes that SMYD4 is available to operate in retinal tissue but does not provide evidence of retinal enrichment.

This is consistent with its Phase 2 classification as a broad regulatory gene rather than a direct retinal candidate.

17. Initial Assessment of the Phase 2 Membrane/Endosomal Cluster

The pre-specified Phase 2 cluster was:

INPP5K + PITPNA + TLCD2 + WDR81 + RILP + SERPINF1

The adult retinal bulk screen gives:

Gene

Retina nTPM

Phase 3A status

SERPINF1

56.9

Strongly supported

PITPNA

43.9

Strongly supported

INPP5K

14.7

Supported

TLCD2

10.1

Supported

WDR81

0.7

Weak in bulk

RILP

0.4

Weak in bulk

The first four genes therefore satisfy the most basic spatial criterion:

their Phase 2 molecular mechanisms are clearly represented in adult human

retinal tissue.

The two endosomal-trafficking genes do not yet satisfy that criterion strongly at the whole-retina level.

Consequently, the Phase 2 convergence is partially supported rather than fully supported by Phase 3A.

That distinction should be preserved.

18. Unexpectedly Strong Early Result: INPP5K–PITPNA–TLCD2–SERPINF1

Before single-cell analysis, four components of the membrane/lipid model already show measurable adult retinal expression:

PITPNA

PI/PC transfer

+

INPP5K

phosphoinositide turnover

+

TLCD2

phospholipid acyl-chain regulation

+

SERPINF1

PEDF-linked retinal lipid signaling.

This provides a preliminary spatial bridge across four molecular levels:

lipid availability → phosphoinositide chemistry → membrane composition →

extracellular retinal lipid signaling.

The result is stronger than Phase 2 alone because these genes are not simply related in abstract molecular databases.

Their transcripts are all present within the same gross tissue compartment:

adult human retina.

However, whole-retina co-presence does not prove they occur in the same cells.

That becomes the central Phase 3C test.

19. PRPF8 Creates an Independent High-Expression Retinal Axis

PRPF8's adult retinal expression is substantially higher than the membrane candidates except SERPINF1/PITPNA.

This means the deletion contains, within the same retinal tissue:

highly expressed RNA-splicing machinery

and: multiple expressed membrane/lipid regulators.

The later coexpression question is therefore particularly important:

Does PRPF8 occupy the same retinal cell classes that express the membrane-homeostasis genes?

If so, there would be spatial opportunity for splicing perturbation and membrane perturbation to coexist within the same cells.

If not, the mechanisms may remain biologically separated.

20. Noncoding Loci

The noncoding loci require separate treatment.

PITPNA-AS1 and MIR22HG

Standard bulk RNA-seq can potentially detect these lncRNAs, but lower abundance, transcript annotation, and library-selection differences make cross-platform absence difficult to interpret.

They will therefore be analyzed through:

- eyeIntegration gene/transcript matrices;
- appropriate retinal single-cell datasets where captured;
- developmental datasets.

MIR22

Mature miR-22 is not validly quantified from conventional poly(A) mRNA RNA-seq.

MIR22 will therefore be excluded from the ordinary mRNA-expression ranking and analyzed later using small-RNA/miRNA-specific datasets.

RN7SL105P

Because RN7SL105P is a highly homologous pseudogene, apparent RNA-seq reads may be ambiguously mapped.

It will not receive an expression score unless locus-specific mapping quality can be established.

21. Current Phase 3A Classification

Strong adult retinal expression

PRPF8

SERPINF1

PITPNA

Clear/moderate adult retinal expression

RPA1

INPP5K

TLCD2

Detectable but lower adult retinal expression

RTN4RL1

SMYD4

Low whole-retina expression

SLC43A2

WDR81

Very low whole-retina expression

SERPINF2

RILP

SCARF1

This classification is based only on the current bulk retinal screen and is not a final biological ranking.

22. Hypothesis Status After Phase 3A

H1 — Membrane/endosomal co-localization

Partially supported.

PITPNA, INPP5K, TLCD2, and SERPINF1 are clearly present in adult retina.

WDR81 and RILP are weak in whole-retina bulk and require cell-type testing.

H2 — Direct retinal localization

Partially supported at bulk level.

PRPF8, SERPINF1, and RTN4RL1 are all detectable in adult retina, with PRPF8 and SERPINF1 particularly strong.

H3 — WDR81/RILP neural-trafficking overlap

Unresolved.

Whole-retina abundance is low for both.

H4 — Developmental enrichment

Not yet tested.

H5 — Magnocellular specificity

Not yet tested.

H6 — Regulatory overlap

Not yet tested at cell-type resolution.

23. Main Interpretation

The first in silico expression layer supports the project enough to continue, but it does not simply confirm the Phase 2 model.

Instead, it modifies it.

The strongest retinal molecular set at this stage is:

PRPF8 + SERPINF1 + PITPNA + RPA1 + INPP5K + TLCD2

with RTN4RL1 also clearly detectable.

The specific Phase 2 membrane convergence is narrowed provisionally to:

PITPNA + INPP5K + TLCD2 + SERPINF1

because those four demonstrate substantial adult retinal expression.

WDR81 and RILP remain highly credible neuronal proteins but are not strongly supported as whole-retina adult mechanisms from this first dataset.

That is a meaningful negative/qualifying result rather than a failure of the hypothesis.

24. Phase 3A Working Conclusion

Phase 3A demonstrates that several of the strongest Phase 2 candidate mechanisms

possess clear spatial opportunity within adult human retina.

Most notably:

PITPNA

INPP5K

TLCD2

and:

SERPINF1

are simultaneously represented in adult retinal tissue, supporting the existence of an ocular membrane/lipid-regulatory subnetwork.

PRPF8 is strongly expressed in the same gross tissue, while RPA1 is also substantially expressed and RTN4RL1 is clearly detectable.

Conversely, WDR81, RILP, SCARF1, SERPINF2, and SLC43A2 show low whole-retina expression, emphasizing that molecular importance identified in Phase 2 does not automatically imply adult retinal importance.

The appropriate conclusion is therefore:

Phase 3A provides preliminary spatial support for a subset of the Phase 2 visual-neural model, especially the PITPNA–INPP5K–TLCD2–SERPINF1 membrane/lipid axis, while leaving the WDR81–RILP trafficking component unresolved and weakening direct retinal prioritization of several systemic genes.

The next analysis must move below the tissue level.

Whole-retina RNA cannot determine whether these genes occur in the same biological cells.

The decisive next question is:

Which retinal cell populations—rods, cones, bipolar cells, amacrine cells, retinal ganglion cells, Müller glia, RPE, vascular cells, or immune cells—express each affected locus?

Phase 3B — Human Ocular Bulk Expression

Editorial scope note: Phase 3B was defined in the frozen Phase 3 protocol but was not

executed as a separate results section before Phase 3C. Its initial ocular bulk screen was incorporated into Phase 3A above. The original Phase 3B protocol text is reproduced verbatim here so that the Phase 3 record remains complete without inventing a standalone result.

GTEx alone is insufficient for detailed eye biology.

The primary ocular bulk resource will be eyeIntegration, which integrates healthy human ocular RNA-seq datasets processed through a consistent pipeline and provides gene- and transcript-level expression matrices. ("eyeIntegration")

The ocular analysis should include, where available:

- neural retina;
- retinal pigment epithelium;
- choroid;
- optic nerve;
- cornea;
- other ocular tissues as internal comparisons.

The main objective is to determine whether the Phase 2 candidates are actually enriched in:

neural retina and/or RPE relative to non-neural ocular tissues.

Phase 3C — Adult Human Retinal Single-Cell Localization

1. Objective

Phase 3A established that several affected genes are detectably expressed in adult human retina at the whole-tissue level.

Phase 3C asks the more discriminating question:

Which adult human retinal cell populations actually express the affected loci, and do the Phase 2 mechanistically related genes occupy the same cellular compartments?

This distinction is essential because bulk retinal RNA-seq can generate apparent convergence even when different genes are expressed in completely different cell populations.

The principal cell classes considered are:

rod photoreceptors, cone photoreceptors, bipolar cells, amacrine cells, horizontal cells, retinal ganglion cells, Müller glia, astrocytes, microglia, and retinal pigment epithelial cells.

RPE is treated separately from the neural retina because it is an epithelial support layer rather than a retinal neuron, despite its critical role in photoreceptor physiology.

2. Reference Data

The principal contemporary reference is the 2026 Human Retina Cell Atlas, which integrated approximately 3.9 million cells from 125 human donors and identified more than **130 retinal cell types. The atlas integrates multiple published studies with approximately 2.7**

million previously unpublished observations and includes both transcriptomic and chromatinaccessibility information. (Li et al.)

For replication and interpretation, older independent human retinal datasets remain useful. The 2019 Lukowski atlas profiled 20,009 cells from three healthy adult donors and resolved all major neural retinal populations, including rods, cones, bipolar cells, amacrine cells, retinal ganglion cells, horizontal cells, Müller glia, astrocytes, and microglia. (Lukowski et al.)

The current gene-level screen uses accessible Human Protein Atlas single-cell annotations together with independent retinal studies. Because some HPA classifications vary between releases and because the largest 2026 atlas is distributed primarily through interactive and downloadable single-cell resources, release-dependent signals are classified as preliminary rather than treated as replicated facts.

3. Evidence Classification

Phase 3C uses four expression-evidence levels.

Level

Interpretation

Reproduced across independent human data or supported

3 — Strong

by current single-cell evidence plus orthogonal retinal
evidence

Convincing localization in one major human single-cell

2 — Moderate

resource

Suggestive, release-dependent, low-abundance, or

1 — Preliminary

indirectly localized

0 — Unsupported

No reliable adult retinal cell-type localization identified

An unsupported result does not mean that the gene is biologically absent. Low
expression, technical dropout, library chemistry, or rare-cell localization may prevent detection.

4. Adult Retinal Cell-Type Matrix

Strongest current cell-type

Locus

Evidence

Phase 3C interpretation

signal

RPE-enriched

INPP5K

RPE

phosphoinositide

mechanism

RPE-enhanced lipid-transfer

PITPNA

RPE

mechanism

Possible glial/immune

SLC43A2

Microglial-associated signal

localization; weak direct

neural-retinal role

Supports immune/vascular

Microglial/astrocytic/

SCARF1

rather than neuronal

endothelial bias

localization

No convincing retina-

Adult retinal cell of origin

RILP

specific enrichment; broader

unresolved

neuronal association

Broad retinal rather than

Broad neural retina,

PRPF8

uniquely photoreceptor-

including rods and cones

specific requirement

Possible cone-associated

TLCD2

Cone photoreceptor signal

1–2

membrane-lipid mechanism

No strong adult retinal

WDR81

Microglial-associated signal

neuronal localization yet

Strongest current cell-type**Locus****Evidence****Phase 3C interpretation****signal**

No meaningful retinal

Supports systemic rather

SERPINF2

specialization identified

than retinal mechanism

Strongly replicated

SERPINF1**RPE**

RPE/PEDF localization

Broad regulatory expression

No retinal-specific

SMYD4

rather than specialized

enrichment

retinal role

Broad/low-specificity

Housekeeping genome-

RPA1

cellular expression

maintenance mechanism

Bulk retinal expression

confirmed, but RGC

RTN4RL1

Cell of origin unresolved

localization not

demonstrated

PITPNA-AS1

Unresolved

—

lncRNA capture limitation

lncRNA/host-transcript

MIR22HG

Unresolved

—

capture limitation

Not validly measurable by

MIR22

—

Requires small-RNA data

standard scRNA-seq

Pseudogene mapping

RN7SL105P

Unresolved

—

ambiguity

5. INPP5K Localizes Strongly Toward Retinal Pigment Epithelium

Current Human Protein Atlas single-cell annotation classifies INPP5K as:

group enriched in retinal pigment epithelial cells and late spermatids

and places it in the expression cluster:

Retinal pigment epithelial cells — Retinol metabolism. ("INPP5K Single Cell")

This substantially improves the Phase 3A interpretation.

INPP5K is not merely present somewhere in whole retinal tissue. Its expression appears preferentially associated with the RPE compartment.

That provides cellular opportunity for its Phase 2 phosphoinositide mechanism to operate at the interface responsible for:

photoreceptor support

↓

outer-segment turnover

↓

retinoid cycling

↓ membrane trafficking and metabolic exchange.

The current evidence does not show that INPP5K loss disrupts these functions. It establishes only that the enzyme is preferentially positioned within an ocular compartment where intensive membrane processing occurs.

Phase 3C status

RPE localization: supported. Photoreceptor localization: not established as the principal signal. RGC localization: not established.

6. PITPNA Also Shows RPE Enhancement

PITPNA independently shows cell-type-enhanced expression in retinal pigment epithelial cells, in addition to several nonocular populations. ("PITPNA Single Cell")

This result is important because PITPNA and INPP5K were already independently linked in Phase 2 through phosphatidylinositol biology.

Their molecular functions differ:

PITPNA

→ phosphatidylinositol/phosphatidylcholine binding and transfer

whereas:

INPP5K

→ phosphoinositide 5-phosphate hydrolysis.

Phase 3C now adds a spatial relationship:

both are preferentially represented in RPE-associated single-cell expression.

This is substantially stronger evidence of functional convergence than their mere presence in the same whole-retina sample.

It still does not demonstrate physical interaction or epistasis.

Phase 3C status

INPP5K + PITPNA RPE co-localization: supported at the cell-class level.

The appropriate term is:

RPE-centered functional convergence.

7. SERPINF1 Produces the Strongest RPE Signal

SERPINF1/PEDF is currently classified as group enriched in retinal pigment epithelial cells and decidual stromal cells in the Human Protein Atlas. ("SERPINF1 Single Cell")

Independent human retinal single-cell work also identified an RPE cluster characterized by high SERPINF1 and RLBP1 expression. (Voigt et al.)

This provides replication across independent human evidence.

SERPINF1 therefore receives: Expression Evidence Level 3 — Strong.

The result aligns closely with its established PEDF biology.

Phase 2 showed that PEDF contributes to:

- retinal trophic support;
- antiangiogenic regulation;
- photoreceptor survival;
- extracellular-matrix signaling;
- PEDF-R/PNPLA2-linked lipid metabolism.

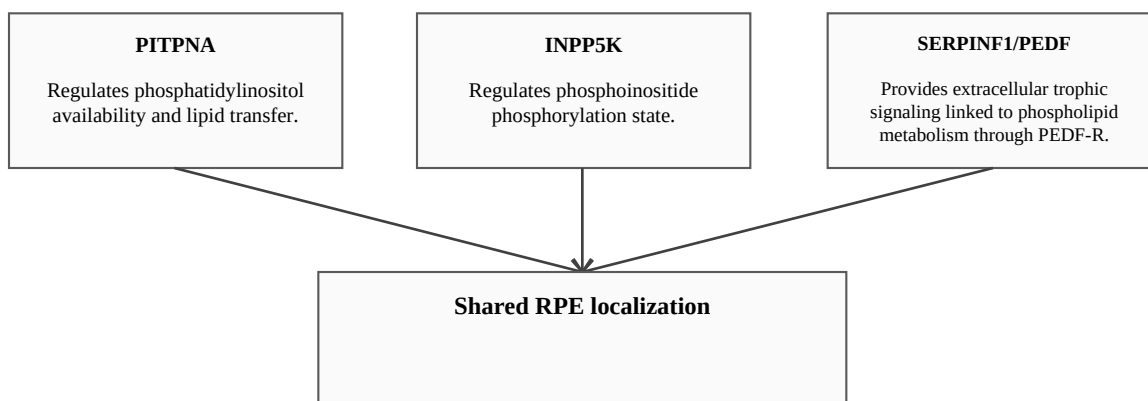
Phase 3C now localizes this mechanism strongly to:

retinal pigment epithelium.

8. An RPE-Centered Molecular Module Now Emerges

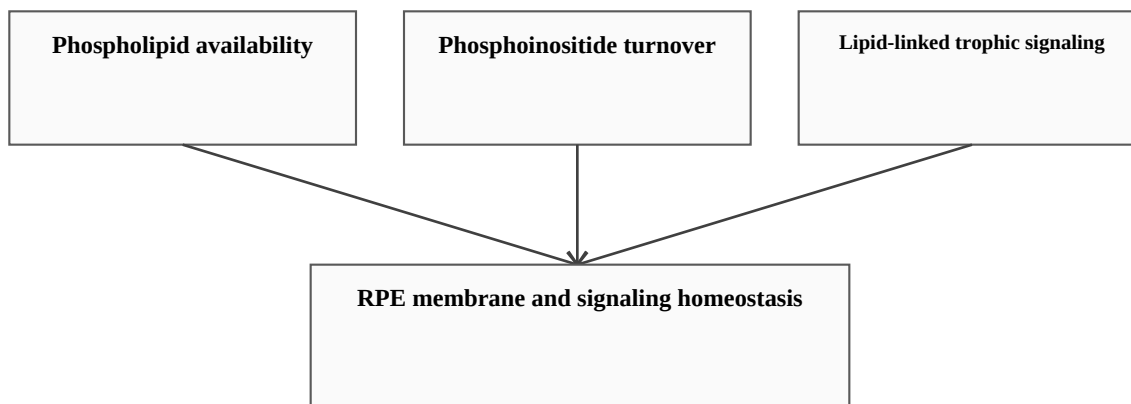
The strongest new Phase 3C result is therefore:

RPE-centered functional convergence



Their functions are chemically different but complementary.

Resulting model



This is currently the strongest cell-type-specific convergence identified anywhere in the project.

Importantly, it differs from the original broad Phase 2 model.

The strongest membrane/lipid convergence in adult retina presently appears to occur primarily at the:

RPE–neural-retina interface

rather than being uniformly distributed throughout retinal neurons.

9. Why the RPE Finding Matters

RPE cells perform unusually membrane-intensive functions.

They support photoreceptors through:

- outer-segment phagocytosis;
- retinoid recycling;
- nutrient transport;
- ion and water regulation;
- growth-factor secretion;
- lipid handling.

Therefore a convergence of lipid and phosphoinositide regulators within RPE is biologically coherent.

However, this observation must not be converted prematurely into:

RPE dysfunction causes the candidate visual phenotype.

At this stage the correct conclusion is narrower:

The deletion simultaneously affects several genes whose normal expression

and molecular functions converge within RPE cells.

That establishes a plausible cellular site of combined dosage perturbation.

10. PRPF8 Is Broadly Expressed Across Neural Retina

Earlier HPA releases classified PRPF8 as enhanced in rod photoreceptor cells and placed it within a photoreceptor/visual-perception expression cluster. ("PRPF8 Single Cell")

However, more recent evidence argues against interpreting PRPF8 as primarily rodspecific.

A 2026 human retinal-organoid study reported PRPF8 expression throughout the neural retina, including both rod and cone photoreceptors, and noted that this distribution is consistent with existing human retinal single-cell data showing broadly uniform retinal expression. (Zimmann et al.)

Therefore the stronger interpretation is:

PRPF8 is broadly required across retinal cell populations, with robust

photoreceptor expression, rather than being a uniquely rod-restricted gene.

This fits its molecular role as a core spliceosomal protein.

Phase 3C status

Adult neural-retinal expression: strong. Rod expression: strong. Cone expression: strong. Strict rod specificity: not supported.

11. PRPF8 Therefore Represents a Different Type of Retinal Risk

The RPE module involves cell-enriched membrane and trophic functions.

PRPF8 is different.

Because it is a core RNA-processing protein expressed broadly through neural retina, reduced PRPF8 dosage—if functionally important—could potentially affect multiple retinal cell populations simultaneously.

Thus:

RPE module

→ spatially concentrated biochemical convergence

whereas:

PRPF8

→ broad retinal information-processing vulnerability.

The two mechanisms could coexist in the same tissue without being part of the same direct pathway.

12. TLCD2 Shows a Possible Cone-Photoreceptor Signal

The Human Protein Atlas v24 classification identifies TLCD2 as cell-type enhanced in **cone photoreceptor cells, along with several nonocular cell populations. ("TLCD2 Single Cell")**

This is potentially important because TLCD2 regulates phospholipid acyl-chain composition and membrane fluidity.

Cone photoreceptors depend on specialized membrane architecture and intense lipid turnover.

However, the TLCD2 signal is currently less secure than the RPE findings because celltype classifications have varied across HPA releases.

Therefore TLCD2 receives:

Expression Evidence Level 1–2: preliminary to moderate.

The current hypothesis is:

TLCD2 dosage loss

↓

possible altered phospholipid composition within cone-associated cells

but not: TLCD2 loss causes cone dysfunction.

Replication in the full 2026 Human Retina Cell Atlas should be required before assigning strong cell-type specificity.

13. The Lipid Cluster Therefore Splits Into at Least Two Adult Retinal Compartments

Phase 3A suggested a broad retinal set:

PITPNA + INPP5K + TLCD2 + SERPINF1.

Phase 3C now indicates that this set is not homogeneous.

The strongest current organization is:

RPE-associated module

PITPNA

INPP5K

SERPINF1

Possible photoreceptor-associated component

TLCD2

particularly:

cone photoreceptors

This distinction is scientifically important.

The deletion may perturb related aspects of membrane biology in adjacent but different retinal compartments rather than through a single intracellular pathway operating in one cell.

14. WDR81 Does Not Show Strong Adult Retinal Neuronal Enrichment

WDR81 is classified by the Human Protein Atlas as cell-type enhanced in microglia, together with several non-neural cell types. ("WDR81 Single Cell")

Phase 3A also found only approximately 0.7 nTPM in whole-retina bulk tissue.

Thus the current adult-retinal evidence does not support strong WDR81 expression in:

- rods;
- cones;
- bipolar neurons;
- amacrine neurons;
- retinal ganglion cells.

The available signal instead suggests a possible microglial association.

This does not weaken WDR81's established neurodevelopmental importance.

It changes where that importance should be investigated.

WDR81 may prove more relevant to:

- developmental neural progenitors;
- brain neurons;
- glial populations;
- dLGN;
- cortical systems

than to mature retinal neurons.

Phase 3C status

Adult retinal-neuronal hypothesis: weakened. Glial/microglial possibility: retained.

Developmental neural hypothesis: still open.

15. RILP Also Fails to Show a Strong Retina-Specific Cell-Type Signal

RILP is associated broadly with a neuronal neurotransmission/signaling expression

cluster, and current HPA data classify it as enhanced in “other brain neurons” among several

non-neural populations. ("RILP Single Cell")

However, no convincing adult-retinal cell-type enrichment emerged from the current screen.

Combined with its very low whole-retina bulk abundance, this argues against making RILP a leading adult retinal mechanism at present.

This is an important negative result.

Phase 2 demonstrated strong neuronal RILP biology.

Phase 3C now suggests that the best place to test that biology may be:

central visual-system neurons rather than mature retina.

Therefore the forthcoming dLGN and V1 analyses become especially important for RILP.

16. The Original WDR81–RILP Retinal Convergence Is Not Supported

Phase 2 predicted that WDR81 and RILP might form a coherent endosomal sequence:

WDR81

→ endosome maturation

↓

RAB7-positive late endosome

↓

RILP

→ organelle transport.

At the molecular level this remains plausible.

At the adult-retina expression level, however, Phase 3C does not currently demonstrate that both genes are strongly expressed in the same retinal neuronal population.

WDR81 shows a microglia-associated signal.

RILP shows a broader neuronal signal without clear retina-specific enrichment.

Therefore:

H3 is not supported in adult retina at this stage.

This does not falsify the molecular pathway.

It means the pathway's spatial opportunity has not yet been demonstrated in mature retinal cells.

It should now be tested in the dLGN, V1, and developmental datasets.

17. RTN4RL1: An Important Negative/Unresolved Finding

Phase 2 identified RTN4RL1 as a strong neural candidate because Nogo-receptor-family

biology directly affects retinal ganglion-cell axons and optic-nerve regeneration.

Phase 3A showed detectable whole-retina expression.

However, current HPA single-cell classification does not identify retinal ganglion cells as an RTN4RL1-enriched population. Instead, the strongest whole-body single-cell specificity appears in spermatogenic cells, while human brain single-nucleus expression shows low overall cell-type specificity. ("RTN4RL1 Single Cell")

Human Protein Atlas protein-level data do support broader neural expression, including neurons, astrocytic/neuropil and microglial/neuropil compartments, but that does not establish a retinal ganglion-cell-specific signal. ("RTN4RL1 HPA")

Therefore the hypothesis:

RTN4RL1 is specifically enriched in adult human retinal ganglion cells

is not currently supported.

This is important because it prevents the Phase 2 optic-nerve literature from being converted into an expression claim that the data do not yet support.

Phase 3C status

Retinal presence: supported by bulk RNA. RGC-specific enrichment: unresolved/not demonstrated. Central neural relevance: still strongly plausible.

18. RPA1 Appears to Behave as a Broad Cellular Maintenance Gene

RPA1 has low overall cell-type specificity in current single-cell resources, with enrichment signals occurring outside the retina, particularly in proliferative or specialized nonretinal populations. Its broader expression pattern is classified as consistent with basic cellular processes. ("RPA1 Single Cell")

This is consistent with RPA1's molecular function.

RPA1 is required for fundamental:

- ssDNA protection;
- replication;
- repair;
- ATR signaling.

Its substantial retinal bulk expression therefore likely reflects a widely distributed **cellular maintenance requirement, rather than specialization within a particular mature retinal**

neuron.

This matters for interpretation.

RPA1 remains the strongest direct heterozygous molecular dosage candidate from Phase 2, but Phase 3C provides no evidence that its dosage vulnerability is retina-specific.

19. SMYD4 Also Shows No Specialized Adult Retinal Localization

SMYD4 currently demonstrates low cell-type specificity overall and no convincing retinal-specific enrichment. Its strongest HPA single-cell enhancement is reported in myonuclei rather than ocular neurons. ("SMYD4 Single Cell")

Thus its Phase 3A retinal bulk expression should be interpreted as:

general expression compatible with a broad regulatory role

rather than:

evidence for a specialized retinal epigenetic mechanism.

The best later opportunity for SMYD4 may be developmental analysis, where epigenetic regulators can have transient effects not apparent in mature tissue.

20. SLC43A2 Shows a Possible Microglial Association

SLC43A2 is not strongly expressed in whole adult retina.

Current single-cell resources nevertheless show an elevated signal involving

microglia/neuropil among several other cell populations, while its overall strongest expression

remains associated with renal and epithelial transport systems. ("SLC43A2 Single Cell")

This supports a limited possibility that SLC43A2 has an immune/glial role within neural tissues.

However, the evidence does not support elevating SLC43A2 into the principal neuralretinal pathway.

Phase 3C status

Photoreceptor role: unsupported. Retinal neuronal role: unsupported. Possible microglial contribution: preliminary.

21. SCARF1 Is Consistent With a Glial/Vascular Rather Than Neuronal Role

SCARF1 shows strong cell-type specificity for several non-neuronal populations, including microglia and astrocytes, and its expression cluster is annotated as brain endothelial cells — blood-brain barrier. ("SCARF1 Single Cell")

This is consistent with its Phase 2 role in:

- scavenger recognition;
- apoptotic-cell clearance;
- immune homeostasis.

Its very low adult-retinal bulk expression makes it unlikely to represent a major photoreceptor or retinal-neuron mechanism.

Instead, if it proves relevant to the visual system, the more plausible route would involve: immune, glial, endothelial, or clearance biology.

22. SERPINF2 Remains Essentially Non-Retinal

SERPINF2 shows strong single-cell enrichment in proximal tubular cells, with liver-associated systemic expression and only very low adult-retinal bulk abundance. ("SERPINF2 Single Cell")

No compelling mature retinal neuronal or RPE cell-type signal emerged.

This strongly reinforces the Phase 2 classification:

SERPINF2 is primarily a systemic hemostatic locus rather than a direct retinal gene.

Its inclusion in later retinal analyses should remain as an unbiased negative control rather than because there is currently a positive retinal hypothesis.

23. Noncoding Loci Require a Separate Assay Strategy

The four noncoding loci cannot be treated like ordinary protein-coding genes.

PITPNA-AS1

As a lncRNA, PITPNA-AS1 may have low abundance and can be poorly captured by droplet-based or 3'-biased scRNA-seq.

Current absence of a strong cell-type annotation is therefore:

unresolved, not evidence of true biological absence.

MIR22HG

The same limitation applies to MIR22HG.

Its host-transcript abundance may be measurable in appropriate total-RNA datasets, but conventional retinal single-cell assays may underrepresent it.

MIR22

Mature miR-22 cannot be validly inferred from ordinary poly(A)-selected scRNA-seq.

Its Phase 3 localization requires: small-RNA sequencing or miRNA-specific expression data.

RN7SL105P

Because RN7SL105P is a homologous pseudogene, short-read mapping may not reliably distinguish it from related 7SL-derived sequences.

No expression score should presently be assigned.

24. Revised Cellular Architecture of the Deletion

Phase 3C now permits the Phase 2/3A model to be reorganized by cellular compartment.

Adult retinal compartment

Most relevant affected loci currently supported

RPE

SERPINF1, INPP5K, PITPNA

Photoreceptors

PRPF8, possible TLCD2 cone enrichment

Broad neural retina

PRPF8, broad RPA1 availability

Microglial/glial-associated

WDR81, SCARF1, possible SLC43A2

Specific RGC enrichment

None demonstrated yet

Adult retinal endosomal WDR81–RILP pair

Not demonstrated

Systemic/non-retinal

SERPINF2 strongly; SLC43A2 predominantly

Unresolved noncoding

PITPNA-AS1, MIR22HG, MIR22, RN7SL105P

This is more biologically informative than the Phase 3A bulk ranking.

25. Hypothesis H1 — Membrane/Endosomal Co-Localization

Original H1 predicted that multiple members of:

INPP5K + PITPNA + TLCD2 + WDR81 + RILP + SERPINF1

would occupy at least one common retinal or visual-neural cell population.

Current result

Partially supported, but substantially revised.

A clear adult RPE module is emerging: INPP5K + PITPNA + SERPINF1.

TLCD2 may instead preferentially involve cones.

WDR81 and RILP are not presently shown to occupy the same adult retinal population.

Therefore the original six-gene same-cell model is:

not supported as a single adult retinal module.

The broader pathway may instead span multiple adjacent cellular compartments.

26. Hypothesis H2 — Direct Retinal Localization

The pre-specified candidates were:

PRPF8

SERPINF1

RTN4RL1.

PRPF8

Strongly supported.

Broad neural-retinal and photoreceptor expression.

SERPINF1

Strongly supported.

Strong replicated RPE localization.

RTN4RL1

Partially supported.

Bulk retinal expression exists, but the specific adult retinal cell population remains unresolved.

Therefore H2 is:

supported for PRPF8 and SERPINF1, unresolved at cell-type level for

RTN4RL1.

27. Hypothesis H3 — WDR81/RILP Neural Trafficking Overlap

In adult retina:

not currently supported.

The two genes do not presently demonstrate convincing shared retinal cell-type enrichment.

This is one of Phase 3C's most important negative results.

The appropriate next test is not to rescue the hypothesis by changing the criterion.

It is to move the same frozen hypothesis into:

human dLGN

and: human V1.

28. Hypothesis H6 — Regulatory Overlap

PRPF8 is broadly present through neural retina and therefore has cellular opportunity to influence transcripts in multiple retinal populations.

The miRNA component cannot yet be evaluated because mature miR-22 requires a different sequencing modality.

Thus: PRPF8 overlap: spatially plausible. MIR22 regulatory overlap: unresolved.

No direct target relationship should yet be inferred.

29. Major New Phase 3C Finding

The principal result is not simply that several deleted genes are expressed in retina. It is that their expression begins to organize into distinct retinal biological compartments.

Strongest current model

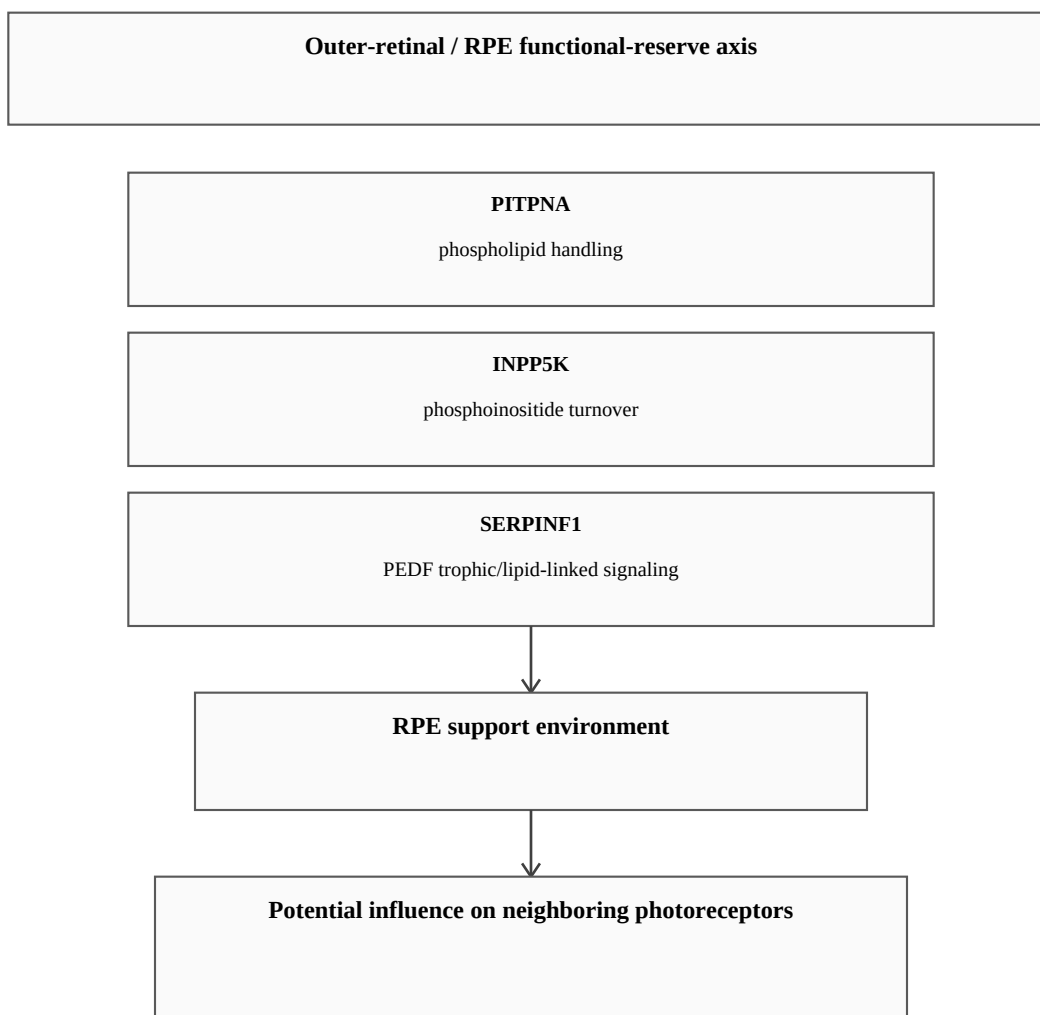
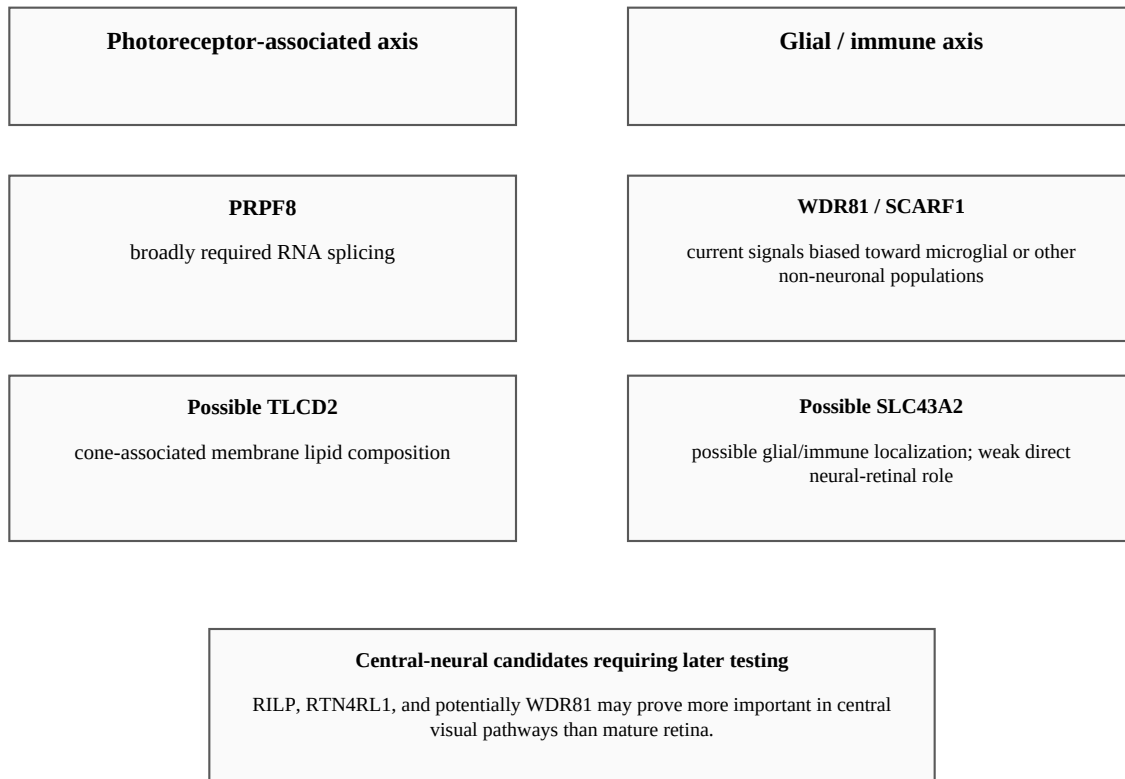


Figure continues on the next page.



WDR81

may prove more important in central visual pathways than mature retina.

30. Implication for the Original Visual-Neurological Question

Phase 3C strengthens the possibility of a retinal biological contribution, but not in the simplistic form initially conceivable.

Current evidence does not suggest:

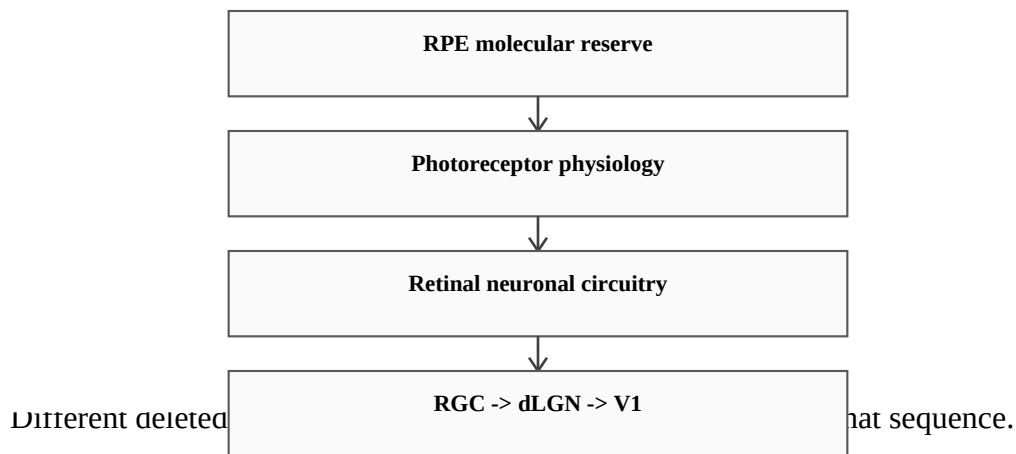
all deleted neural genes are simultaneously weakened inside the same retinal neuron.

Instead it suggests a distributed system:

RPE molecular reserve

↓

Distributed visual-system sequence



This is potentially more biologically realistic than a single-cell, single-pathway model.

It also makes later visual-pathway analysis essential.

31. Important Negative Findings

Phase 3C currently provides no strong evidence for:

RTN4RL1-specific enrichment in retinal ganglion cells.

It provides no strong evidence that: WDR81 and RILP form a coexpressed adult-retinal trafficking module.

It provides no evidence that: SERPINF2 is a significant retinal gene.

It provides no convincing specialized retinal localization for: SMYD4.

And conventional scRNA-seq cannot resolve: mature MIR22 dosage.

These negative results should be retained in the frozen dataset rather than discarded.

32. Current Cell-Type Priority Ranking

For adult retinal biology specifically, the strongest current candidates are:

Rank class**Loci****Reason****SERPINF1, INPP5K, PITPNA,**

Strong cell-type localization plus

Tier A**PRPF8**

established molecular relevance

Relevant expression but weaker retinal

Tier B**TLCD2, RPA1**

specificity or mechanism

Strong prior biology but adult retinal

Tier C**RTN4RL1, WDR81, RILP**

cell localization remains unresolved

Weak/specialized non-neuronal retinal

Tier D**SCARF1, SLC43A2, SMYD4**

evidence

Little evidence for direct adult retinal

Tier E**SERPINF2**

participation

PITPNA-AS1, MIR22HG, MIR22,

Separate assay required

RNA-class/technical limitations

RN7SL105P

This ranking is only for adult retinal localization.

It does not replace the Phase 2 molecular-priority ranking.

33. Evidence Boundary

Phase 3C now permits statements such as:

INPP5K, PITPNA and SERPINF1 show convergent expression within the

RPE compartment.

It permits:

PRPF8 is broadly expressed across human neural retina, including

photoreceptors.

It tentatively permits:

TLCD2 may be enhanced in cone photoreceptors.

It does not permit:

The deletion impairs RPE function.

It does not permit:

The deletion damages cones.

It does not permit:

The RPE module causes visual stress.

And it does not permit:

The genes interact epistatically.

Expression establishes:

cellular opportunity for a mechanism.

It does not establish pathological execution of that mechanism.

34. Phase 3C Working Conclusion

Adult human retinal single-cell analysis substantially refines the bulk-expression model developed in Phase 3A.

The strongest finding is a reproducible RPE-centered convergence involving

SERPINF1, INPP5K, and PITPNA. These genes independently regulate extracellular retinal

trophic signaling, phosphoinositide metabolism, and phospholipid handling, respectively.

Their shared RPE localization provides the first strong evidence that multiple deleted genes with related membrane/lipid functions occupy the same biologically relevant human ocular cell compartment.

PRPF8 represents a second major retinal axis. Rather than being narrowly restricted to rods, current evidence supports broad expression through the neural retina, including rods and cones, consistent with a fundamental spliceosomal function.

TLCD2 provides a preliminary cone-photoreceptor signal that may extend the membranelipid mechanism into neural retina, but replication is required.

Conversely, the expected WDR81–RILP endosomal pair is not currently supported as a shared adult retinal module. WDR81 shows a more microglial-associated expression pattern,

whereas RILP lacks strong retina-specific enrichment and retains a broader neuronal profile. RTN4RL1 also lacks demonstrated retinal ganglion-cell enrichment despite its strong experimental optic-nerve biology.

Phase 3C therefore supports retinal involvement while narrowing and reorganizing

the mechanism.

The strongest present model is no longer:

one six-gene membrane pathway operating uniformly throughout retina.

It is:

an RPE-centered membrane/phosphoinositide/trophic module, a separate photoreceptor-associated RNA/lipid module, and additional neural-trafficking genes whose principal relevance may emerge farther downstream

in the central visual pathway or during development.

That distinction should now be carried forward unchanged into the next analyses.

Phase 3D — Human Dorsal Lateral Geniculate Nucleus Analysis

1. Objective

Phase 3D tests whether the loci affected by the 17p13.3 deletion show preferential

localization within one of the principal human dorsal lateral geniculate nucleus visual relay systems:

magnocellular — M

parvocellular — P

or: koniocellular — K.

This analysis directly addresses one of the project's pre-specified exploratory hypotheses:

H5 — One or more affected loci may show preferential expression in human

magnocellular dLGN neurons.

The analysis is deliberately symmetric.

Evidence favoring P or K neurons would be retained with equal weight, and absence of pathway-specific enrichment is treated as a meaningful negative result.

2. Biological Context

The dorsal lateral geniculate nucleus is the principal thalamic relay between:

retina

↓

dLGN

↓ primary visual cortex — V1.

In primates, visual information has traditionally been divided into magnocellular, parvocellular and koniocellular streams.

These populations differ substantially in:

- morphology;
- receptive-field properties;
- temporal response;
- contrast sensitivity;
- chromatic processing;
- retinal input;
- cortical projection patterns.

The transcriptomic question is whether those physiological distinctions are accompanied by correspondingly strong differences in gene expression.

Current human evidence indicates that the answer is only partly yes.

3. Primary Human Dataset

The principal dataset is the Allen Institute/Bakken et al. Comparative LGN study.

The Allen Institute currently provides processed human LGN expression and metadata for 1,209 nuclei. ("Comparative LGN")

The published analysis used approximately 950 neuronal nuclei from three postmortem

human donors to construct the principal neuronal taxonomy. The study identified:

- three GABAergic neuronal populations;
- two koniocellular populations;
- one combined magnocellular/parvocellular projection-neuron population.

The two koniocellular populations were designated:

Kap

and: Kp.

The remaining major glutamatergic population was designated:

MP. (Bakken et al.)

This point is central to interpretation.

4. Human M and P Neurons Do Not Form Cleanly Separated Transcriptomic Classes

Despite the well-established physiological distinction between magnocellular and parvocellular neurons, human M and P nuclei did not resolve into two clean transcriptomic clusters.

Instead:

M and P neurons largely formed a common MP cluster.

Within that cluster, the investigators identified a continuous M-to-P expression gradient.

Cross-species integration with macaque data confirmed that the human ends of this continuum correspond to more clearly separable macaque M and P populations. (Bakken et al.)

Therefore:

M versus P

in adult human dLGN is better represented transcriptomically as:

a graded molecular distinction

rather than:

two strongly separated gene-expression states.

This places an important constraint on the original magnocellular hypothesis.

5. Published Human M-versus-P Marker Test

The study identified 32 genes showing consistent M-versus-P enrichment across all three human donors.

Validated examples included:

Magnocellular-associated

CRH

SUSD2

with:

ROBO2

also marking the M-directed end of the conserved human/macaque continuum.

Parvocellular-associated

EPHA7

FOXP2.

A pan-M/P marker included:

BTNL9. (Bakken et al.)

The complete published Figure 3 marker set includes genes such as:

- ABCC3
- ADAMTSL3
- AKAP2
- ARHGAP6
- ARHGEF28
- CDH6
- CDH8
- CNKSR3
- COL11A1
- COL21A1
- CRH
- EGFEM1P
- ELL2
- FAM19A5
- FER1L6
- FOXP2
- GALNTL6
- GPC5
- GRID2
- HPSE2
- KCNH7

- KCNJ6
- LUZP2
- MPPED2
- PLCL1
- PRKG1
- PTER
- ROBO2
- RUNX1
- SH3RF2
- SPON1
- SUSP2
- TMTC1
- TOX
- VSNL1
- EPHA7

across the donor-specific and joint M/P comparisons displayed by the study. (Bakken et al.)

6. None of the 17 Deleted Loci Is a Published M/P Marker

The affected loci are:

****INPP5K**

PITPNA

SLC43A2

SCARF1

RILP

PRPF8

TLCD2

WDR81

SERPINF2

SERPINF1

SMYD4

RPA1

RTN4RL1

PITPNA-AS1

MIR22HG

MIR22

RN7SL105P.**

None appears among the published genes showing the consistent human M-versus-P marker pattern.

Therefore:

No affected locus currently demonstrates published evidence of preferential

magnocellular transcriptional enrichment in the adult human dLGN.

Equally:

No affected locus emerges as a strong published parvocellular marker.

This is a genuine negative result for the pre-specified analysis.

7. This Does Not Mean the Genes Are Absent From M or P Neurons

The distinction is critical.

A gene can be expressed in:

both M and P neurons

without being: differentially expressed between M and P neurons.

For example, a housekeeping or broadly neuronal protein could be essential to both populations while never appearing in an M-versus-P marker list.

Therefore the result supports:

no demonstrated M-specific enrichment

but does not support:

no expression in magnocellular neurons.

Those are different propositions.

8. Magnocellular Hypothesis Status

The frozen Phase 3 hypothesis was:

One or more affected genes may show preferential expression in human magnocellular dLGN neurons.

The current result is: H5 — Not supported at the published marker level.

None of the affected loci appears among the consistent human M-enriched markers.

Therefore the project should not currently claim that the deletion preferentially targets the magnocellular pathway.

This substantially narrows the original hypothesis.

9. Why This Result Is Particularly Important

A weaker analytical strategy could have proceeded from:

visual stress

↓

historical magnocellular hypotheses

↓

search for any deleted neural gene

↓

claim magnocellular involvement.

Phase 3D does not permit that inference.

When the actual human M/P transcriptomic dataset is examined first, the affected genes do not appear among the genes distinguishing M from P neurons.

Thus:

current transcriptomic evidence does not support a magnocellular-specific

mechanism for this deletion.

This is one of the clearest examples so far of the project's phenotype-neutral design protecting against confirmation bias.

10. Koniocellular Analysis

Human dLGN transcriptomics resolves the K populations more clearly than M and P.

The Bakken et al. study identified:

Kap

and:

Kp

koniocellular types.

The principal reported K-associated markers include:

CALB1

CAMK2A

PRKCG

with:

PENK

selectively identifying the Kp population.

PENK-positive Kp neurons were anatomically confirmed in posterior K1 and K2 layers using RNA in situ hybridization. (Bakken et al.)

11. No Deleted Locus Is a Defining K Marker

None of the 17 affected loci is among the primary K-defining markers reported by the human study.

Thus the current evidence also fails to support:

selective koniocellular involvement.

At the transcriptomic-marker level, the deletion does not map specifically onto:

M

P

or:

K

identity.

12. Gene-by-Gene dLGN Specificity Screen

Published M-specific

Published P-specific

Affected locus

Defining K marker

Current interpretation

marker

marker

No relay-class

INPP5K

No

No

No

specificity demonstrated

No relay-class

PITPNA

No

No

No

specificity demonstrated

No relay-class

SLC43A2

No

No

No

specificity demonstrated

No relay-class

SCARF1

No

No

No

specificity demonstrated

Neural relevance

RILP

No

No

No

retained; M/P/K

specificity unsupported

Likely broader cellular

PRPF8

No

No

No

role rather than relay-

class marker

No relay-class

TLCD2

No

No

No

specificity demonstrated

Neural relevance

WDR81

No

No

No

retained; M/P/K

specificity unsupported

No dLGN specialization

SERPINF2

No

No

No

demonstrated

Strong retinal role does

SERPINF1

No

No

No

not extend to dLGN

subtype specificity

No relay-class

SMYD4

No

No

No

specificity demonstrated

Consistent with broad

RPA1

No

No

No

cellular maintenance

role

Neural signaling

RTN4RL1

No

No

No

relevance retained; no

M/P/K specialization

PITPNA-AS1

No evidence

No evidence

No evidence

Unresolved

MIR22HG

No evidence

No evidence

No evidence

Unresolved

Published M-specific**Published P-specific****Affected locus****Defining K marker****Current interpretation****marker****marker**

Not appropriately

Not appropriately

Not appropriately

MIR22

measured as mature

Separate assay required

measured

measured

miRNA

Mapping/function

RN7SL105P

No evidence

No evidence

No evidence

unresolved

The table should be interpreted as a specificity matrix, not an absolute-expression matrix.

13. RILP

RILP was specifically important entering Phase 3D because Phase 3C failed to establish a strong adult retinal localization despite compelling neuronal trafficking biology.

The dLGN analysis does not rescue RILP as a magnocellular-specific candidate.

RILP is not among the published:

- M markers;
- P markers;
- K markers.

Thus its neuronal trafficking mechanism may still operate in visual-relay neurons, but current evidence does not associate it preferentially with one dLGN visual channel.

This leaves RILP as a:

general neural-trafficking candidate rather than a magnocellular candidate.

14. WDR81

The same conclusion applies to WDR81.

Phase 2 established strong neural-development and endosomal PI3P biology.

Phase 3C suggested that adult retinal expression may be more glial than neuronal.

Phase 3D finds no evidence that WDR81 is a defining:

- magnocellular;
- parvocellular;
- koniocellular

transcriptomic marker.

Therefore:

WDR81 remains a strong neural mechanism but not a visual-relay-class-

specific mechanism.

Its strongest remaining Phase 3 opportunity may lie in:

- developmental brain;
- broader thalamic neurons;
- cortical populations.

15. RTN4RL1

RTN4RL1 was another important candidate because Phase 2 established direct axonal and optic-nerve relevance.

However, RTN4RL1 also fails to appear among the published M/P/K defining gene sets.

Therefore its functions involving:

- axonal growth;
- CSPG recognition;

- synaptic structural regulation

should currently be interpreted as:

general neuronal structural biology

rather than evidence of a magnocellular pathway specialization.

This is consistent with Phase 3C, where retinal ganglion-cell-specific enrichment was also not demonstrated.

Two independent Phase 3 analyses therefore now fail to support a highly specialized RTN4RL1 visual-channel localization.

16. PRPF8

PRPF8 likewise shows no M/P/K marker specificity.

This is biologically unsurprising.

Its function is fundamental spliceosomal RNA processing.

Phase 3C demonstrated broad neural-retinal availability rather than narrow retinal subtype restriction.

Phase 3D now supports the same general interpretation centrally:

PRPF8 is more plausibly a broadly required cellular factor than a transcriptomic marker of a specific visual relay stream.

This does not weaken PRPF8's retinal relevance.

It weakens only a hypothesis of magnocellular-selective PRPF8 vulnerability.

17. PITPNA, INPP5K, TLCD2 and SERPINF1

The Phase 3C retinal lipid module does not reappear as an M-specific dLGN transcriptional module.

None of:

PITPNA

INPP5K

TLCD2

or:

SERPINF1

is among the published human M/P discriminating genes.

Therefore the strongest Phase 3C finding remains principally:

retinal/RPE biology

rather than:

magnocellular thalamic biology.

This begins to spatially separate the two hypotheses.

18. A Critical Property of the Human dLGN Dataset

Failure to find M-specific enrichment should not be exaggerated.

The original study itself found that human M and P neurons are remarkably similar transcriptomically relative to their major physiological and morphological differences.

Only a limited set of genes consistently distinguishes the two.

The human M/P cells remain one MP transcriptomic population with a molecular gradient rather than completely distinct cell types. (Bakken et al.)

Consequently:

physiological specialization need not require large differences in steady-state

gene expression.

A deleted gene expressed equally in M and P neurons could theoretically have different functional consequences if the two neuronal types impose different physiological demands.

That possibility remains open.

However, there is currently no positive transcriptomic evidence supporting it.

19. Independent Primate Evidence Supports the Complexity of LGN Cell Identity

A 2026 reanalysis of the Allen macaque LGN dataset identified additional heterogeneity within the conventional M/P/K categories:

- two magnocellular subpopulations;
- two parvocellular subpopulations;
- three koniocellular subpopulations.

The analysis used 1,309 macaque LGN nuclei and found substantial within-class transcriptomic complexity. (Sun, Renshaw, and Pezaris)

This reinforces an important conceptual point:

M, P and K are themselves not molecularly homogeneous populations.

The recent macaque analysis does not highlight RTN4RL1, RILP, WDR81, PRPF8, PITPNA, INPP5K, SERPINF1, TLCD2, or RPA1 among the principal subtype genes discussed in the article. (Sun, Renshaw, and Pezaris)

This is supporting primate evidence only.

It does not replace the human analysis.

20. Revised Interpretation of the Magnocellular Hypothesis

At the start of the project, one conceivable mechanism was:

17p13.3 deletion

↓

specific disruption of genes important to the magnocellular pathway

↓

altered temporal/contrast processing

↓

candidate visual-stress phenotype.

Phase 3D does not support the critical middle step.

The current evidence instead favors:

17p13.3 deletion

↓

multiple retinal and broadly neuronal molecular perturbations

↓

possible visual-system vulnerability

without evidence of: preferential magnocellular transcriptional targeting.

This is a substantially more conservative and defensible model.

21. Consequence for the Visual-Stress Hypothesis

The historical literature around visual stress has sometimes invoked magnocellular dysfunction.

Phase 3D means that this project should not use that literature as a shortcut to connect

the CNV with visual stress.

At present:

The genomic findings do not independently predict a magnocellular-specific

lesion.

If later physiological evidence were to implicate magnocellular processing, that would require a separate explanation.

The present transcriptomic evidence does not supply it.

22. What Phase 3D Does Support

Phase 3D does support a different conclusion.

Several of the affected genes have substantial:

- neural;
- retinal;
- membrane;
- trafficking;
- signaling

functions.

But those mechanisms appear more likely to operate as:

broad cellular systems

than as molecular identities unique to one thalamic visual channel.

This is especially consistent with:

PRPF8

→ universal spliceosomal machinery

RPA1

→ genome-maintenance machinery

RILP

→ general neuronal endosomal transport

WDR81

→ general endosomal signaling

RTN4RL1

→ broadly neuronal structural signaling.

23. Cross-Phase Comparison

Phase 3 has now produced three distinct spatial findings.

Phase 3A — Whole Retina

Strong expression of:

****PRPF8**

SERPINF1

PITPNA

RPA1

INPP5K

TLCD2**

with weaker adult-retinal signals for several trafficking genes.

Phase 3C — Retinal Cell Types

Strongest convergence:

INPP5K + PITPNA + SERPINF1

within the:

RPE compartment

with PRPF8 broadly available through neural retina and possible TLCD2 cone association.

Phase 3D — dLGN

No affected locus is currently identified as a strong:

M

P

or:

K

marker.

Thus the most specific spatial convergence discovered so far remains:

RPE-centered rather than magnocellular-centered.

24. Hypothesis Status After Phase 3D**H1 — Membrane/endosomal colocalization**

Partially supported, principally through RPE rather than dLGN subtype specialization.

H2 — Direct retinal localization

Supported for PRPF8 and SERPINF1; strong RPE convergence with INPP5K/PITPNA.

H3 — WDR81/RILP neural-trafficking overlap

Not demonstrated in adult retina and not demonstrated as M/P/K-specific in human

dLGN.

Still open in cortical/developmental analyses.

H4 — Developmental enrichment

Not yet tested.

H5 — Magnocellular specificity

Not supported.

This is now a meaningful negative result.

H6 — Regulatory overlap

Still unresolved.

25. Evidence Boundary

Phase 3D permits the statement:

None of the 17 affected loci has emerged as a published defining marker of

adult human magnocellular, parvocellular, or koniocellular dLGN identity.

It also permits:

The available human transcriptomic data do not support preferential

magnocellular targeting by the deletion.

It does not permit:

The affected genes are absent from magnocellular neurons.

Nor does it permit:

Magnocellular physiology cannot be affected.

Those stronger claims would require direct quantitative expression and physiological data.

26. Technical Limitation

The Allen Institute makes the complete processed human LGN expression matrix and metadata publicly downloadable, containing 1,209 nuclei. ("Comparative LGN")

The published paper and associated supplementary analyses provide the marker-level M/P/K results used here, including the 32 reproducible human M/P genes and the defining K markers. The underlying processed dataset is also deposited under GSE182211 and through the Allen Comparative LGN resource. ("GSE182211"; "Comparative LGN")

A final frozen Phase 3 matrix should ultimately include direct numerical extraction of each affected gene's expression across individual M, P and K nuclei.

Therefore the present conclusion is appropriately classified as:

strong marker-level negative evidence for subtype specificity

rather than:

complete quantitative absence of differential expression.

27. Phase 3D Working Conclusion

Human dLGN transcriptomics does not currently support the hypothesis that the 17p13.3 microdeletion preferentially disrupts a magnocellular molecular program.

Human magnocellular and parvocellular relay neurons themselves are transcriptionally much more similar than their well-known physiological differences might suggest. They form a largely shared MP transcriptomic population with a comparatively subtle M-to-P expression gradient. A limited set of genes—including CRH, SUSP2 and ROBO2 toward the magnocellular end and EPHA7 and FOXP2 toward the parvocellular end—defines this gradient. None of the 17 loci affected by the index deletion appears among the published reproducible M/P marker genes. (Bakken et al.)

Likewise, the affected loci do not overlap the principal human koniocellular markers CALB1, CAMK2A, PRKCG and PENK.

Accordingly:

The pre-specified magnocellular-specificity hypothesis is not supported by

current adult human dLGN transcriptomic evidence.

This result does not weaken the broader proposition that the deletion may affect visualneural biology. Instead, it changes the likely mechanism.

The strongest spatial evidence remains the RPE-centered INPP5K–PITPNA–

SERPINF1 convergence, combined with broad retinal PRPF8 expression and additional neural

mechanisms that do not appear restricted to a particular thalamic visual stream.

The current model should therefore shift from:

a magnocellular-specific defect

toward:

a distributed retinal and neural functional-reserve model in which several molecular systems may be altered without preferentially targeting one

classical M/P/K visual channel.

That is a stronger conclusion scientifically because it follows the transcriptomic evidence rather than the original phenotype hypothesis.

Phase 3C - Integrated Expression and Spatiotemporal Analysis

Research Scope and Status

Phase 3 evaluates where and when genes disrupted by the GRCh37/hg19

chr17:1,411,408-1,841,103 microdeletion are expressed, with particular emphasis on the retina, retinal pigment epithelium (RPE), dorsal lateral geniculate nucleus (dLGN), primary visual cortex (V1), and developing human neocortex. The phase does not treat expression as evidence of disease causation. Its purpose is narrower: to determine whether genes already identified as structurally disrupted and mechanistically plausible in Phases 1 and 2 have spatial and temporal opportunity to influence visual-neural biology.

The deletion interval contains thirteen protein-coding loci considered in the Phase 3 expression analysis (INPP5K, PITPNA, SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, RPA1, and RTN4RL1) and four noncoding annotations

(PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P). Phase 3 preserves the structural interpretation established previously: the deletion begins within INPP5K and ends within RTN4RL1, with the latter losing essentially the canonical coding sequence on the affected allele. Those structural facts are background constraints rather than expression results.

Abstract

Phase 3 integrates bulk tissue expression, single-cell and single-nucleus transcriptomics,

retinal cell-type resources, human dLGN transcriptomics, adult human brain atlases, developmental cortical multiomics, and spatial fetal-cortex transcriptomics. The strongest direct cell-localized convergence remains retinal and RPE-centered: INPP5K and PITPNA show RPE-associated expression, while SERPINF1 is strongly associated with retinal pigment epithelial cells and has an established retinal trophic context. Human Protein Atlas data classify INPP5K as group-enriched in retinal pigment epithelial cells and PITPNA as cell-type enhanced in retinal pigment epithelial cells; SERPINF1 is likewise group-enriched in retinal pigment epithelial cells (Human Protein Atlas, “INPP5K”; Human Protein Atlas, “PITPNA”; Human Protein Atlas, “SERPINF1”). PRPF8 is broadly expressed in retina rather than restricted to one visual cell class, TLCD2 has a cone-photoreceptor signal that remains preliminary for mechanistic interpretation, WDR81 is more compatible with microglial/non-specific cellular expression than a mature retinal neuronal module, and RTN4RL1 is detectably expressed in human retina but has not been demonstrated as retinal-ganglion-cell-specific (Human Protein Atlas, “TLCD2”; Human Protein Atlas, “WDR81”; Human Protein Atlas, “RTN4RL1”).

Central visual-system analysis does not support a magnocellular-specific model. Singlecell and single-nucleus profiling of primate dLGN found that human magnocellular and parvocellular projection neurons are transcriptomically similar relative to their marked

physiological differences; the affected 17p13.3 loci do not emerge as defining published M-, P-, or K-cell markers (Bakken et al.). Adult cortical analysis similarly supports distributed neural expression rather than a deletion-wide V1-specific signature. Developmental analysis strengthens evidence that selected genes have temporal opportunity during cortical development, especially WDR81, RTN4RL1, RILP, and SERPINF1, but independent human developmental atlases demonstrate readily detectable V1 specialization without identifying the deletion interval as a conspicuous V1-defining program (Bhaduri et al.; Wang et al.; Qian et al.). The resulting model is therefore one of distributed functional reserve: several deleted genes plausibly constrain membrane, phosphoinositide, endosomal, trophic, RNA-processing, or genome-maintenance systems at different visual-system sites and developmental stages, but the complete causal chain from heterozygous dosage loss to biochemical disturbance to visual-neural phenotype has not been demonstrated.

Methods and Evidentiary Rules

Phase 3 was designed as an expression-opportunity analysis rather than a phenotype-

association screen. Evidence was interpreted in the following order: genomic disruption, known molecular function, tissue or cell-type expression, spatiotemporal overlap, pathway convergence, perturbational evidence, phenotype association, and causality last. Accordingly, detection of a transcript in retina or cortex establishes only that the gene is biologically available in that compartment. It does not establish that the heterozygous deletion changes transcript abundance, protein abundance, enzymatic flux, synaptic physiology, or perception.

Adult tissue-level observations were treated as lower-resolution evidence than cell-resolved data. Single-cell and single-nucleus findings were prioritized when the relevant cell type was represented and assay limitations were understood. Developmental comparisons were

not interpreted as longitudinal trajectories unless the underlying study design supported that inference. Cross-sectional BrainSpan or Allen-derived standardized expression values from different donors and ages were therefore treated as suggestive temporal observations rather than within-subject fold changes. BrainSpan provides RNA-seq and microarray expression data across multiple cortical and subcortical structures and developmental periods, including prenatal laser-microdissection resources (Allen Institute for Brain Science).

Noncoding loci required additional caution. Standard polyadenylated or nuclear RNA workflows are not reliable assays for mature microRNA abundance; consequently, MIR22 cannot be considered absent or low merely because it is poorly represented in ordinary singlenucleus RNA-seq. MIR22HG and MIR22 were also treated as related host/microRNA annotations rather than independent deletion events. PITPNA-AS1 and RN7SL105P remain technically underdetermined in the available visual-system single-cell resources.

Evidence labels used throughout this phase are: established, when supported directly by the relevant human dataset or well-validated cell-resolved resource; suggestive, when supported by cross-sectional, lower-resolution, or indirect evidence; inference, when a mechanistic interpretation combines independent observations without direct experimental demonstration; and unsupported, when the proposed specificity or relationship is not demonstrated by the reviewed evidence.

Phase 3A: Adult Whole-Tissue and Ocular Expression

The initial adult expression screen asked whether the affected genes were detectably expressed in retina and brain before attempting cell-type localization. This screen supported substantial adult retinal expression opportunity for PRPF8, SERPINF1, PITPNA, RPA1, INPP5K, and TLCD2, with lower or less discriminatory whole-retina signals for several other

interval genes. Whole-tissue expression was not treated as evidence that all of these genes act in the same retinal cell. The Human Protein Atlas retina resource illustrates the interpretive problem: the retina expresses a large fraction of the human proteome, so bulk detection alone has limited specificity (Human Protein Atlas, “Retina Proteome”).

RTN4RL1 is a useful example. Current Human Protein Atlas data report detectable retinal RNA, with a GTEx-derived consensus around 7.9 nTPM, supporting retinal opportunity but not assigning that expression to retinal ganglion cells or another specific neuronal population (Human Protein Atlas, “RTN4RL1”). TLCD2 likewise shows retinal RNA in bulk data, while its single-cell annotation includes cone photoreceptor enhancement; this makes cone involvement a reasonable candidate for follow-up, not an established cone-specific deletion mechanism (Human Protein Atlas, “TLCD2”).

Phase 3A therefore established a permissive rather than causal result: multiple affected

genes are present in adult ocular tissue, so retinal mechanisms cannot be excluded on expression grounds. The phase then moved to cell-resolved localization because bulk retinal abundance could not distinguish photoreceptor, RPE, glial, vascular, immune, or inner-retinal contributions.

Phase 3B-3C: Retinal Cell Types and RPE-Centered Convergence

Human retinal single-cell resources provide a substantially stronger anatomical test. Cowan and colleagues sequenced 285,441 single cells from retinal organoids across developmental time points and from adult human peripheral retina, fovea, pigment epithelium, and choroid, demonstrating that retinal disease-related genes can show marked cell-type specificity (Cowan et al.). Independent adult retinal atlases likewise resolve foveal and peripheral

retinal cell populations at single-cell scale (Yan et al.). These datasets make it possible to ask whether mechanistically related deletion genes actually occupy compatible retinal compartments.

The strongest Phase 3 retinal result is the RPE-centered convergence of INPP5K, PITPNA, and SERPINF1. Human Protein Atlas single-cell annotations classify INPP5K as group-enriched in retinal pigment epithelial cells and place it in an RPE-associated expression cluster. PITPNA is cell-type enhanced in RPE, while SERPINF1 is group-enriched in RPE and decidual stromal cells (Human Protein Atlas, “INPP5K”; Human Protein Atlas, “PITPNA”; Human Protein Atlas, “SERPINF1”). This is important because the three genes are not merely co-expressed names: their known molecular functions occupy complementary membrane, lipid, phosphoinositide, and trophic biology. Phase 3 does not claim that they form a single biochemical complex or that their heterozygous loss produces a measured RPE defect. It establishes spatial opportunity for functional convergence.

PRPF8 presents a different pattern. It is broadly expressed in retina and is biologically relevant to retinal disease under established variant mechanisms, but its expression is not restricted to one RPE or photoreceptor subtype. Consequently, PRPF8 is best treated as a broadly available RNA-processing constraint rather than evidence for the RPE membrane module. Its presence may matter if future perturbational work demonstrates altered splicing in affected visual cells, but expression alone cannot supply that missing step.

TLCD2 provides a weaker but potentially informative signal. Current Human Protein Atlas single-cell annotation lists cone photoreceptor cells among TLCD2-enhanced cell types, while bulk retina RNA is detectable (Human Protein Atlas, “TLCD2”). This supports a preliminary cone-associated expression opportunity. It does not establish that TLCD2 dosage

affects cone membrane composition in the deletion carrier, nor that such an effect contributes to photophobia, pattern glare, color processing, or another visual phenotype.

WDR81 did not produce the expected mature retinal neuronal convergence with RILP. Human Protein Atlas annotations describe WDR81 as having low overall cell-type specificity with enhancement in microglia among several cell types (Human Protein Atlas, “WDR81”). RILP, by contrast, is assigned to a neuronal neurotransmission-and-signaling expression cluster in the broader single-cell atlas (Human Protein Atlas, “RILP”). These findings do not demonstrate a shared adult retinal neuronal trafficking module. The WDR81-RILP relationship therefore remained mechanistically plausible from Phase 2 but anatomically unresolved in adult retina.

The retinal conclusion is correspondingly asymmetric. INPP5K-PITPNA-SERPINF1 represents the strongest direct cell-localized convergence in Phase 3. PRPF8 contributes broad retinal RNA-processing opportunity; TLCD2 has preliminary cone-associated support; RTN4RL1 is expressed in retina but lacks demonstrated retinal-ganglion-cell enrichment; and WDR81-RILP co-localization in mature retinal neurons has not been established.

Phase 3D: Human dLGN and the Magnocellular Hypothesis

Phase 3D tested whether the deletion genes preferentially map to the magnocellular

visual pathway. This was a particularly important falsification test because a magnocellular model could otherwise become an attractive but weakly constrained explanation for visual discomfort or contrast-related symptoms. Bakken and colleagues profiled primate dLGN using single-cell and single-nucleus RNA sequencing, including nuclei from anatomically defined human magnocellular, parvocellular, and koniocellular regions. They identified four principal thalamocortical neuron types in primates but found relatively few strong transcriptional

differences between M and P populations despite their well-known morphological and physiological differences (Bakken et al.).

In human dLGN, M and P neurons largely occupy a shared transcriptomic MP cluster with a gradient rather than two completely separate molecular classes. Published M-associated markers include CRH and SUSD2, whereas EPHA7 is among validated P-associated markers; the study also identifies K-cell classes separately (Bakken et al.). None of the affected 17p13.3 loci emerged as a defining published M, P, or principal K marker in this analysis.

This is a negative result with a specific meaning. It does not show that the deletion genes are absent from magnocellular neurons. It shows that the available human dLGN transcriptomic evidence does not support preferential magnocellular enrichment. Because M and P neurons are themselves transcriptomically similar, absence from a marker list cannot be treated as proof of equal functional importance in both populations. Nevertheless, the data fail to support a deletionwide magnocellular-specific mechanism. Hypothesis H5 is therefore classified as unsupported.

Phase 3E: Adult Human V1 and Central Cortical Distribution

Phase 3E extended the analysis from the thalamus to human cerebral cortex, with

particular interest in V1. Large adult human brain atlases demonstrate extensive neuronal and glial diversity across the brain and provide a framework for distinguishing broadly expressed genes from regionally specialized programs. Siletti and colleagues profiled more than three million nuclei from approximately one hundred dissections and identified thousands of transcriptomic subclusters, including area-specific cortical neurons (Siletti et al.).

Within this framework, several affected genes have credible central-neural expression, but the interval does not emerge as a coherent adult V1-specific program. RILP gains support as

a neuronal gene because the Human Protein Atlas places it in a neuronal neurotransmission- and signaling expression cluster (Human Protein Atlas, “RILP”). RTN4RL1 has human cerebral cortex and retinal expression and is classified as tissue enriched in cerebral cortex in current Human Protein Atlas protein-summary metadata, but this does not establish V1 or cortical-layer specificity (Human Protein Atlas, “RTN4RL1”). PITPNA and INPP5K remain broadly available rather than demonstrably V1-selective.

SERPINF1 retains trophic and vascular/glial opportunities in central tissue but does not reproduce the same sharply localized RPE pattern in V1. WDR81 and RILP remain compatible with endosomal and neuronal trafficking biology, yet Phase 3E found no evidence that WDR81-RILP-RTN4RL1 form a synchronized adult V1 cell-type module. PRPF8 and RPA1 are even less regionally discriminatory because their core functions in RNA processing and genome maintenance make broad expression biologically expected.

The adult cortical result therefore favors distributed vulnerability over a simple visual cortex lesion model. The genes can be biologically active in cortex without being selectively enriched in V1, layer 4, intratelencephalic neurons, corticothalamic neurons, or another single cortical class. This distinction became the rationale for Phase 3F: if adult specificity was weak, developmental timing might still reveal a period during which the deleted genes are unusually relevant to visual-cortex formation.

Phase 3F: Human Visual-Cortex Development and Developmental Enrichment

Phase 3F initially centered on the developmental multiomic atlas of Wang and colleagues

and was subsequently broadened to independent cortical arealization and spatial-transcriptomic resources. Wang et al. analyzed 27 brain specimens and 38 biological samples spanning five developmental stages from the first trimester through adolescence. Paired single-

nucleus RNAseq and ATAC-seq were generated from PFC and V1, producing 243,535 nuclei after initial quality control and a final cortical dataset of approximately 232,000 nuclei. The study resolved progenitor, neuronal, glial, immune, and vascular classes and explicitly modeled regional developmental differences (Wang et al.).

The atlas is particularly valuable because it identifies a V1-specific layer-4 intratelencephalic trajectory rather than assuming that all layer-4 neurons are equivalent. Common and V1-specific EN-L4-IT populations share an early trajectory and diverge later in development. The authors reported 1,908 differentially expressed genes distinguishing V1-specific from common EN-L4-IT neurons and highlighted markers such as CUX1 and KCNIP1. Their analysis used sample-level pseudobulk logic with age and dataset effects and Benjamini-Hochberg multiple-testing correction (Wang et al.). None of the affected 17p13.3 loci is among the conspicuous published markers defining that V1-specific branch.

A crucial evidence boundary was maintained throughout the Table 18 investigation. The supplementary archive containing the full differential-expression table is publicly listed, but the exact per-gene rows for all affected loci were not successfully retrieved in the research environment. Consequently, Phase 3 does not label the deletion genes statistically nonsignificant in that comparison. The justified statement is narrower: they are not among the highlighted defining V1-specific EN-L4-IT markers, while their exact effect sizes and adjusted P values remain an archival verification item.

The broadened search materially strengthens the interpretation because independent datasets show that V1 specialization is readily detectable when present. Bhaduri and colleagues profiled ten major brain structures and six neocortical areas during peak neurogenesis and early gliogenesis. They found extensive area-specific differential expression in the second trimester,

including in radial glia, with areal signatures becoming more pronounced as cells differentiate toward excitatory neurons. Their results support strong frontal-versus-occipital molecular patterning rather than a cortex that is transcriptionally homogeneous during development (Bhaduri et al.).

Qian and colleagues provide an even stronger spatial constraint. Using MERFISH and complementary approaches, they constructed a spatial atlas exceeding 18 million fetal cortical cells across eight cortical areas and seven developmental time points. They detected a discrete V1-V2 molecular boundary by approximately gestational week 20 and found early synaptogenesis-related changes in V1-specific layer-4 neurons (Qian et al.). Thus, the available technology and datasets are capable of detecting visual-cortex-specific developmental programs. The failure of the 17p13.3 interval to emerge as a conspicuous defining program cannot simply be attributed to an inability of developmental transcriptomics to resolve V1. Developmental Candidate: WDR81

WDR81 remains one of the strongest early-development candidates in the interval, but the evidence supports developmental opportunity rather than V1 specificity. Cross-sectional developing-brain resources suggest prenatal cortical expression, and its established endosomal biology gives that expression mechanistic relevance. However, WDR81 is not identified as a major V1 arealization determinant in the Bhaduri, Wang, or Qian studies. The Phase 3F classification is therefore: developmental neural relevance, suggestive to moderate; preferential V1 specification, unsupported. Developmental Candidate: RTN4RL1

RTN4RL1 is detectably expressed in human retina and brain and has a neuronal cell surface biology compatible with neurite and circuit maturation. BrainSpan and Allen developmental resources provide temporal expression opportunity, while current Human Protein

Atlas data show cerebral-cortex enrichment at the tissue level (Allen Institute for Brain Science; Human Protein Atlas, “RTN4RL1”). The expression pattern is better described as developmentally dynamic than as a simple prenatal peak. Because RTN4RL1 also suffers nearcomplete canonical coding loss on the affected allele, it remains a high-priority mechanistic candidate. Yet it is not established as a V1-specific arealization gene. Developmental Candidate: RILP

RILP retains a credible neuronal trafficking role but has weaker evidence for early cortical specification. Its neuronal expression-cluster assignment supports central-neural opportunity (Human Protein Atlas, “RILP”), while the developmental atlases do not identify it as a major V1 arealization or layer-4 determinant. The most conservative interpretation is that RILP may matter during neuronal maturation and endolysosomal cargo transport rather than during initial specification of V1 identity. Developmental Candidate: SERPINF1

SERPINF1 bridges the strongest retinal result and the developmental-cortical analysis. Its RPE enrichment is direct, while a 2026 developmental-brain analysis reported a rapid increase in SERPINF1 expression between approximately 24 gestational weeks and four months after birth (Musante et al.). This observation is secondary to that study's primary disease question, so it should not be over-weighted; nevertheless, it supports a developmental trophic opportunity during the late-prenatal to early-postnatal transition. It does not establish V1 specificity.

Functional Ordering without a Demonstrated Molecular Chain

The broadened developmental evidence permits a cautious temporal functional model: WDR81 is most compatible with earlier endosomal and developmental regulation; RTN4RL1 with later structural, neurite, and circuit-maturation opportunity; and RILP with neuronal endolysosomal transport during maturation. This ordering is an inference assembled from

independent functional and expression observations. It is not evidence that one gene activates the next, that the genes form a shared transcriptional module, or that heterozygous loss produces a sequential biochemical cascade.

Wang et al. also inferred hundreds of enhancer-associated regulatory programs from paired RNA and chromatin-accessibility data. The presence of a mapped regulatory landscape makes a shared deletion-specific V1 regulon testable in principle. At present, however, no published result establishes that the affected genes constitute a coordinated V1-specific eRegulon. Regulatory hypothesis H6 therefore remains technically testable but unsupported at the deletion-module level.

Integrated Gene-Level Interpretation

INPP5K. Strong RPE localization opportunity; phosphoinositide biology makes the RPE finding mechanistically coherent. Developmental cortical expression is compatible with a broader role, but V1 specificity is not demonstrated. Evidence status: established RPE expression opportunity; developmental regional specificity unsupported.

PITPNA. RPE-enhanced expression and neuronal/synaptic biology support membranelipid opportunity in retina and brain. The strongest Phase 3 use of PITPNA is as part of the RPE-centered convergence, not as a V1 marker. Evidence status: established RPE opportunity; V1 specificity unsupported.

SLC43A2. Expression opportunity is broad and not strongly visual-system-specific in the reviewed Phase 3 resources. Evidence status: low discriminatory value for Phase 3.

SCARF1. The available expression pattern is more compatible with immune/vascular biology than a specialized retinal-neuronal or V1 program. Evidence status: low support for a direct visual-neuronal localization hypothesis.

RILP. Central neuronal expression is supported, and its trafficking function remains mechanistically relevant. Adult retinal neuronal co-localization with WDR81 and developmental V1 specificity are not demonstrated. Evidence status: moderate neuronal opportunity; specific visual module unresolved.

PRPF8. Broad retinal and neural expression is expected for a core spliceosomal factor. It remains biologically important but spatially nondiscriminatory. Evidence status: strong expression opportunity; low regional specificity.

TLCD2. Bulk retinal expression and cone-photoreceptor enhancement provide a preliminary cell-type lead. Evidence status: suggestive cone-associated opportunity; mechanistic and dosage consequences untested.

WDR81. Developmental and central-neural opportunity is supported, with microglial enhancement in current single-cell annotations. It remains a strong developmental candidate because of independent endosomal biology, not because it is a V1-specific marker. Evidence status: moderate developmental relevance; V1 specificity unsupported.

SERPINF2. Phase 3 provides little evidence for a specialized visual-neural expression pattern. Evidence status: low visual-system discriminatory value.

SERPINF1. Strong RPE enrichment and late-prenatal/early-postnatal developmental expression dynamics make SERPINF1 one of the most spatially informative genes in the interval. Evidence status: strong retinal/RPE opportunity; developmental trophic opportunity supported; V1 specificity unsupported.

SMYD4. Broad/low-specificity expression provides little anatomical discrimination. Evidence status: low Phase 3 specificity.

RPA1. Broad expression is consistent with genome-maintenance function. Its importance is more likely to emerge through dosage-sensitive cellular stress than visual-system localization. Evidence status: strong general cellular opportunity; low visual specificity.

RTN4RL1. Retinal and cortical expression plus severe allelic coding disruption make RTN4RL1 a high-priority neuronal candidate. Cell-surface and circuit-maturation biology are compatible with visual-neural relevance, but retinal-ganglion-cell, V1-layer, and magnocellular specificity remain unproven. Evidence status: moderate-to-high neural opportunity; regional specificity unsupported.

PITPNA-AS1 / MIR22HG / MIR22 / RN7SL105P. Available Phase 3 assays are inadequate for confident visual-system localization of these noncoding loci, especially mature MIR22. Evidence status: technically underdetermined; no negative inference from ordinary snRNA-seq is permitted.

Hypothesis Adjudication

H1: Membrane/endosomal colocalization. Partially supported. The strongest cellocalized evidence is the RPE-centered INPP5K-PITPNA-SERPINF1 configuration. A comparable deletion-wide neuronal module has not been demonstrated in retina, dLGN, or V1.

H2: Direct retinal localization. Supported for selected genes, especially INPP5K, PITPNA, SERPINF1, PRPF8, TLCD2, and RTN4RL1 at differing levels of cellular resolution. Direct localization does not establish pathological effect.

H3: WDR81-RILP trafficking overlap. Mechanistically plausible but anatomically unresolved. Adult retinal neuronal co-localization is not demonstrated; developmental evidence is more compatible with temporally distributed roles.

H4: Developmental enrichment. Partially supported. Several genes have credible developmental cortical opportunity, but the interval is not demonstrated as a V1-specific developmental program.

H5: Magnocellular specificity. Unsupported. Human dLGN transcriptomics does not identify the affected loci as defining magnocellular markers, and no later developmental evidence restores this hypothesis.

H6: Shared V1 regulatory overlap. Presently unsupported. Modern multiomic atlases make the question testable, but no shared affected-gene V1-specific eRegulon has been established.

Interpretive Synthesis: Developmental and Cellular Functional Reserve

The most defensible lay and technical synthesis of Phase 3 is a functional-reserve model. Several genes in the deletion are expressed in biologically relevant visual or neural compartments, and some occupy complementary molecular systems. In RPE, the convergence is unusually concrete because INPP5K, PITPNA, and SERPINF1 share cell-type opportunity while contributing to phosphoinositide, lipid/membrane, and trophic biology. In the developing cortex, WDR81, RTN4RL1, RILP, SERPINF1, PRPF8, and RPA1 provide different forms of developmental or cellular opportunity, but not a single V1-specific transcriptional program.

The term functional reserve is preferable to assuming a fixed fifty-percent loss of function. A heterozygous deletion removes one genomic copy, but the downstream protein or pathway effect depends on transcriptional compensation, protein stability, feedback regulation, redundant pathways, cell type, developmental stage, and physiological demand. Phase 3 therefore supports the proposition that the deletion could reduce reserve in several molecular systems; it does not quantify that reduction.

Likewise, the project has not yet demonstrated a complete chemical or molecular chain linking the deletion to a visual-neural phenotype. Individual biochemical functions are known: INPP5K acts in phosphoinositide metabolism; PIP2A participates in phosphatidylinositol/phosphatidylcholine transfer; WDR81 and RILP participate in endosomal/lysosomal trafficking; SERPINF1/PEDF has trophic and lipid-linked retinal biology; PRPF8 is a core spliceosomal component; RPA1 supports replication and genome maintenance; and RTN4RL1 is a neuronal surface receptor-family protein. What remains unproven is the connecting chain from heterozygous dosage reduction to altered pathway chemistry in the relevant human visual cell, then to altered circuit function, and finally to a reproducible phenotype.

Negative Findings and Falsification Value

Phase 3 generated several negative findings that materially constrain the research. First,

the deletion does not presently support a magnocellular-specific model. Second, adult V1 does not reveal a deletion-wide region- or layer-specific expression program. Third, the mature retinal WDR81-RILP trafficking pair was not demonstrated as a shared neuronal module. Fourth, the developmental interval genes do not emerge among the conspicuous V1-specific layer-4 markers in modern human developmental atlases. Fifth, no shared V1-specific regulatory network among the affected genes has been established.

These negative findings are not failures of the project. They reduce the number of plausible mechanistic models and prevent the analysis from drifting toward a preferred phenotype narrative. The strongest surviving model is more distributed: retinal/RPE membranotropic vulnerability combined with central-neural developmental, trafficking, RNA-

processing, and genome-maintenance constraints. This model is less anatomically simple but better aligned with the evidence.

Limitations

The principal limitation is that expression is not perturbation. None of the reviewed atlases directly measures the molecular state of cells carrying this specific heterozygous deletion. Normal-reference expression establishes opportunity, not dosage response. A gene may be highly expressed yet fully compensated after heterozygous loss, or modestly expressed yet functionally dose-sensitive.

Second, cell-resolved atlases have sampling and technical limitations. Single-nucleus methods can underrepresent cytoplasmic transcripts, rare cell populations can be missed, and regional dissections are imperfect. The dLGN study itself notes practical limitations in isolating thin K layers (Bakken et al.). Spatial assays solve some localization problems but often measure restricted gene panels or snapshots rather than complete developmental trajectories.

Third, developmental datasets are generally cross-sectional across different donors. Apparent age-related changes should not be interpreted as within-person trajectories unless the analysis explicitly models age across samples. The older Allen/BrainSpan signals used in Phase 3F were therefore treated as supportive context, not definitive temporal kinetics.

Fourth, the full gene-level statistics from Wang et al. Supplementary Table 18 remain incompletely extracted for the affected loci. This prevents a formal statement that each gene is statistically nonsignificant in the V1-specific versus common EN-L4-IT comparison. The correct record is that none is among the conspicuous published defining markers and exact per-gene statistics remain pending archival verification.

Finally, the available evidence does not justify attributing Irlen Syndrome, visual stress, photophobia, pattern glare, cortical hyperexcitability, epilepsy, retinal degeneration, or another clinical phenotype to this deletion. If later phases address such phenomena, they should be decomposed into measurable traits and tested against molecular and physiological data rather than treated as a single genetically established syndrome.

Phase 3 Final Conclusion

Phase 3 establishes that the 17p13.3 deletion intersects multiple genes with credible

spatial and temporal opportunity in the human visual system. The strongest direct anatomical convergence is RPE-centered and involves INPP5K, PITPNA, and SERPINF1. Additional genes contribute broader retinal, neuronal, cortical, developmental, or cellular-maintenance opportunity. Human dLGN data do not support preferential magnocellular enrichment. Adult cortical data favor distributed rather than V1-specific expression. Developmental multiomic and spatial atlases demonstrate that V1-specific programs are detectable, yet the affected interval does not presently emerge as a major defining V1 developmental module.

The resulting evidence supports moderate mechanistic plausibility for a distributed reduction in visual-neural developmental and cellular functional reserve. It does not establish a direct dosage-to-chemistry-to-circuit-to-phenotype chain. Phase 4 should therefore shift from expression opportunity toward perturbational evidence: whether heterozygous reduction of the highest-priority genes measurably changes phosphoinositide handling, membrane composition, endosomal trafficking, trophic signaling, RNA processing, genome-stress responses, neurite/circuit maturation, or combinations of these processes in relevant human cellular models.

Phase 3 Frozen Record

Phase 3 may be considered substantively complete subject to one archival refinement:

retrieval of exact affected-gene rows from Wang et al. Supplementary Table 18. That unresolved lookup should remain recorded as a quantitative limitation rather than converted into a negative biological result. The principal conclusions are sufficiently replicated across independent retinal, dLGN, adult-brain, developmental multiomic, and spatial-transcriptomic resources to support transition to Phase 4.